MINIT MESYUARAT KE-99
JAWATANKUASA KECIL PTJ - SUMBER MANUSIA (JKPTJSM)
PUSAT PENGAJIAN TEKNOLOGI INDUSTRI, UNIVERSITI SAINS MALAYSIA

Tarikh	**:**	**17 Julai 2025, Khamis**
Masa	**:**	**9:00 pagi**
Tempat	**:**	**Bilik Persidangan, Pusat Pengajian Teknologi Industri**

Hadir :

Pengerusi
Profesor Madya Dr. Yusri bin Yusup
Dekan, Pusat Pengajian Teknologi Industri

Ahli Jawatankuasa

Timbalan Dekan
Profesor Madya Dr. Uthumporn Utra @ Sapina Abdullah
Timbalan Dekan (Akademik, Kerjaya dan Antarabangsa)
Pusat Pengajian Teknologi Industri

Ahli Majlis
YBhg. Profesor Dato' Dr. Azhar bin Mat Easa
Pusat Pengajian Teknologi Industri

Profesor Madya Dr. Mardiana Idayu binti Ahmad
Pusat Pengajian Teknologi Industri

Panel Luar – Kelompok Sains
Profesor Dr. Mohamad Nasir bin Mohamad Ibrahim
Pusat Pengajian Sains Kimia

Setiausaha / Wakil Pendaftar
Puan Nooraida binti Mad Naser
Timbalan Pendaftar, Pusat Pengajian Teknologi Industri

Jemputan :

Profesor Madya Dr. Lee Chee Keong
Timbalan Dekan (Penyelidikan, Inovasi & Libatsama Industri-Komuniti)
Pusat Pengajian Teknologi Industri
(jam 12:15 tengah hari – 12:30 tengah hari)

Ts Dr. Siti Baidurah binti Yusoff
Pengerusi Rancangan
Bahagian Teknologi Bioproses
(jam 9:00 pagi – 11:50 pagi)

Ts. Dr. Azniwati binti Abd Aziz
Pengerusi Rancangan
Bahagian Teknologi Biosumber
(jam 11:50 pagi – 12:10 tengah hari)

Dr. Mohamad Hafizi bin Abu Bakar
Pusat Pengajian Teknologi Industri
(jam 12:15 tengah hari – 12:30 tengah hari)

MIN 1 : UCAPAN ALUAN PENGERUSI

Pengerusi memulakan mesyuarat dengan mengucapkan terima kasih kepada semua ahli khasnya Profesor Dr. Mohamad Nasir bin Mohamad Ibrahim dari Pusat Pengajian Sains Kimia selaku Ahli Panel Luar PTJ.

Seterusnya Pengerusi memaklumkan bahawa tujuan mesyuarat ini adalah untuk mempertimbangkan empat (4) perkara berikut:

1. Pelantikan Baharu Staf Akademik (Kontrak-Bukan Warganegara / Ekspatriat)
2. Pengesahan Dalam Jawatan
3. Pelantikan Semula Staf Akademik Kontrak Warganegara Pensyarah Universiti DS13 bagi Tujuan Memenuhi Syarat Lantikan Tetap
4. Pelantikan Felo Adjung

MIN 2 : PELANTIKAN BAHARU STAF AKADEMIK (KONTRAK-BUKAN WARGANEGARA / EKSPATRIAT)

Berikut adalah maklumat ringkas mengenai pelantikan ini:

1. Iklan kekosongan jawatan:

   a. Satu (1) slot lantikan jawatan Pensyarah Universiti (Kontrak – Bukan Warganegara/ Ekspatriat) yang diluluskan oleh Universiti melalui surat bertarikh 14 April 2025; USM. 65/9 Jld. CXLIII bagi menggantikan seorang Pensyarah Universiti (Kontrak – Bukan Warganegara/Ekspatriat) yang tamat perkhidmatan:

Bil	Bidang	Slot Lulus	Iklan
1.	Bioprocess Engineering / Biotechnology Engineering / Chemical Engineering / Environmental Engineering or Engineering Technology (Applied in Computer-Aided Design / Separation and Biological Treatment).	1	Lampiran 1A

   Staf yang telah tamat perkhidmatan:

Bil	Nama Staf	Tarikh Lantikan	Tamat Kontrak
1.	Dr. Esam Bashir Abdulsalam Yahya	01.09.2022	23.04.2025

   b. Tarikh Iklan: 25 April 2025 – 26 Mei 2025

2. Ringkasan statistik pencalonan:

Bil	Negara	Jumlah	Tidak Layak	Untuk Saringan 2	Senarai Pendek	Temu Duga
1	Malaysia	3	3	0		
2	Nigeria	11	1	10		
3	Indonesia	2		2	2 (1 tarik diri)	1
4	Iran	3		3		
5	Pakistan	6		6	2	2
6	India	7		7	1	1
7	Yemen	1		1		
8	Sri Lanka	1		1		
9	Bangladesh	1		1		
10	Libya	1		1		
		36	**4**	**32**	**5**	**4**

3. Calon – senarai pendek dan jemputan temu duga

Bil	Calon	Warganegara	Lampiran
1	Dr. Ts. Irvan Dahlan Mansyoer	Indonesia	Lampiran A
2	Dr. Muhammad Zubair Shahid	Pakistan	Lampiran B
3	Dr. Padmesh Turuneveli Narayanapillai	India	Lampiran C
4	Dr. Zakir Khan	Pakistan	Lampiran D

Keputusan:

Jawatankuasa Kecil PTJ – Sumber Manusia (JKPTJSM) Pusat Pengajian Teknologi Industri telah bersidang dan mempertimbangkan perkara berikut untuk dibentangkan dalam Mesyuarat Jawatankuasa Pemilih Kumpulan Pengurusan dan Profesional (Akademik):

Perkara: Pelantikan Baharu Staf Akademik (Kontrak-Bukan Warganegara / Ekspatriat)

Bil	Calon	Warganegara	Keputusan	Lampiran
1	Dr. Zakir Khan	Pakistan	Diperakukan - Pilihan Pertama	Lampiran A
2	Dr. Ts. Irvan Dahlan Mansyoer	Indonesia	Diperakukan - Pilihan kedua dan tertakluk kepada semakan status calon - Sekiranya tidak sesuai, calon pilihan ketiga akan dimajukan untuk pertimbangan Jawatankuasa Pemilih.	Lampiran B
3	Dr. Muhammad Zubair Shahid	Pakistan	Diperakukan - Pilihan ketiga (calon simpanan)	Lampiran C
4	Dr. Padmesh Turuneveli Narayanapillai	India	Tidak diperakukan	Lampiran D

MIN 3 : PENGESAHAN JAWATAN

Jawatankuasa dimaklumkan bahawa Staf Tetap (Percubaan) berikut telah pun memenuhi syarat pengesahan jawatan staf akademik:

1. Dr. Ana Masara Binti Ahmad Mokhtar (Warganegara)

A. Syarat Umum Pengesahan Jawatan Staf Akademik

- Lulus Program Transformasi Minda
- Lulus Modul Pengajaran & Pembelajaran anjuran Pusat Pembangunan Kecemerlangan Akademik (CDAE)
- Lulus Modul Penyeliaan berstruktur anjuran Pusat Pembangunan Kecemerlangan Akademik (CDAE)
- Lulus Bahasa Malaysia Peringkat Maju (Ujian Lisan, Ujian Bertulis dan Ujian Penyampaian Kuliah)
- Menyempurnakan pengajaran sekurang-kurangnya dua (2) semester dengan melengkapkan semua dokumentasi kursus-kursus yang terlibat
- Diperakukan oleh Ketua Jabatan
- Markah LPP berdasarkan purata melebihi :
  a. Tahun 1 : 85%
  b. Purata Tahun 1 dan Tahun 2 : 80%
  c. Purata Tahun 1, Tahun 2 dan Tahun 3: 80 %
  d. Tempoh Pelanjutan I = 80% (tahun semasa)
  e. Tempoh Pelanjutan II = 80% (tahun semasa)

B. Syarat PTJ Bidang Kejuruteraan dan Teknologi

Menghasilkan penerbitan (WoS/Scopus/Buku)	Menyelia pelajar di peringkat Master / PhD	Geran penyelidikan	Lain-lain syarat
Tiga (3) Penerbitan sebagai *Corresponding Author / First Author* dalam WoS **ATAU** Satu (1) artikel dalam Jurnal berindeks WoS sebagai *Corresponding Author / First Author* **DAN** dua (2) bab dalam buku penyelidikan berindeks WoS/Scopus/MAPIM **ATAU** Satu (1) artikel dalam Jurnal berindeks WoS sebagai *Corresponding Author / First Author* **DAN** satu (1) buku penyelidikan berindeks WoS/Scopus/MyCITE. **Nota**: Penerbitan adalah dalam bidang kepakaran staf akademik.	Menyelia satu (1) pelajar PhD sebagai Penyelia Bersama (sekurang-kurangnya 12 bulan) **DAN** Telah menggraduatkan dua (2) pelajar siswazah Mod Campuran/Kerja kursus MSc sebagai Penyelia Utama **ATAU** Menyelia satu (1) pelajar siswazah MSc/PhD (Penyelidikan) sebagai Penyelia Utama (sekurang-kurangnya enam (6) bulan bagi MSc dan 12 bulan bagi PhD) **ATAU** Menyelia dua (2) pelajar Mod Campuran sebagai Penyelia Utama.	Telah mendapat Geran dalaman sebagai Penyelidik Utama. **DAN** Telah memohon satu (1) Geran luar (Kebangsaan/Industri / Antarabangsa) sebagai Penyelidik Utama.	Menghantar borang permohonan jurutera professional **DAN** Khidmat atau sumbangan kepada PTJ atau Universiti.

Sumber: Hebahan Pendaftar 9/2024 23 Januari 2024

Keputusan:

Jawatankuasa Kecil PTJ – Sumber Manusia (JKPTJSM) Pusat Pengajian Teknologi Industri telah bersidang dan mempertimbangkan pengesahan dalam jawatan staf akademik berikut untuk dibentangkan dalam Mesyuarat Jawatankuasa Pemilih Kumpulan Pengurusan dan Profesional (Akademik):

BIL	CALON	LAMPIRAN
1	Dr. Ana Masara Binti Ahmad Mokhtar (Warganegara)	Lampiran E

MIN 4 : PELANTIKAN SEMULA STAF AKADEMIK KONTRAK WARGANEGARA PENSYARAH UNIVERSITI DS13 BAGI TUJUAN MEMENUHI SYARAT LANTIKAN TETAP.

Jawatankuasa dimaklumkan mengenai Hebahan Pendaftar Bil. 6/2024 bertarikh 23 Januari 2024 dan Hebahan Pendaftar Bil. 52/2024 bertarikh 16 Julai mengenai penetapan syarat lantikan tetap staf akademik. Lanjutan daripada penetapan syarat lantikan tetap bagi staf akademik kontrak Warganegara dan calon baharu, pihak Universiti telah menambah baik penetapan syarat lantikan tetap jawatan Pensyarah Universiti DS51, Pensyarah Universiti DS53 (Profesor Madya) dan Pensyarah Universiti Gred Khas C VK7 (Profesor) seperti berikut:

1. Penetapan Syarat Lantikan Tetap Staf Akademik
   a. Pensyarah Sedia Ada
      i. Berkhidmat di USM secara lantikan kontrak sekurang-kurangnya dua (2) tahun pada 31.12.2023
      ii. Mencapai purata markah Laporan Penilaian Prestasi (LPP) 85%
      iii. Menerbitkan sekurang-kurangnya empat (4) penerbitan dalam jurnal berindeks (WoS / Scopus) sebagai First Author / Corresponding Author.
      iv. Pengalaman pengajaran sekurang-kurangnya satu (1) semester.
      v. Diperakukan oleh Ketua Jabatan.
      vi. Lantikan ke jawatan tetap bagi yang layak berkuat kuasa pada 01.01.2024.

   b. Pensyarah Baharu
      i. Menerbitkan sekurang-kurangnya empat (4) penerbitan dalam jurnal berindeks (WoS/Scopus) sebagai First Author/ Corresponding Author.
      ii. Pengalaman pengajaran sekurang-kurangnya satu (1) semester / pengalaman dalam industri yang berkaitan minimum tiga (3) tahun.
      iii. Diperakukan oleh Ketua Jabatan.

2. Penambahbaikan Syarat Lantikan Kontrak Ke Tetap
   a. Staf akademik kontrak sedia ada diberikan tempoh satu (1) tahun lantikan semula kontrak bagi tujuan memenuhi syarat lantikan tetap. Kegagalan memenuhi syarat lantikan tetap menyebabkan staf berkenaan tidak dilantik semula selepas tamat tempoh kontrak satu (1) tahun tersebut.

   b. Tarikh kuat kuasa pelantikan tetap bagi staf yang telah memenuhi syarat lantikan tetap dan berkhidmat dalam lantikan kontrak sekurang-kurangnya dua (2) tahun pada 31 Disember 2023 adalah pada 1 Januari 2024.

3. Satu (1) calon PPTI:

   1. Dr. Sumarni Binti Mansur (Warganegara)

Keputusan:

Jawatankuasa Kecil PTJ – Sumber Manusia (JKPTJSM) Pusat Pengajian Teknologi Industri telah bersidang dan memperakukan pelantikan semula staf akademik kontrak warganegara Pensyarah Universiti DS13 berikut bagi tujuan memenuhi syarat lantikan tetap staf akademik untuk dibentangkan dalam Mesyuarat Jawatankuasa Pemilih Kumpulan Pengurusan dan Profesional (Akademik):

BIL	CALON	LAMPIRAN
1	Dr. Sumarni Binti Mansur (Warganegara)	Lampiran F

MIN. 5 : **Pelantikan Felo Adjung**

Pusat pengajian melalui Mesyuarat Jawatankuasa Pengurusan Pusat Pengajian Teknologi Industri Bil. 2/2025 pada 11 Julai 2025 telah mempertimbangkan permohonan / pencalonan Felo Adjung berikut:

BIL	CALON	WARGANEGARA	TEMPOH	LAMPIRAN
1	Dr. Quamrul Hassan (Bukan Warganegara) Centre for Global Initiatives, Osaka University, Osaka, Japan.	Japan	Setahun	Lampiran G

Berikut adalah Penetapan Baharu dan Sasaran Lantikan Akademik Contract for Service (CFS) bagi Felo Adjung (Hebahan Pendaftar Bil. 12/2025 bertarikh 17 Februari 2025)

Definisi	Lantikan individu yang masih aktif / dalam perkhidmatan dan mempunyai kepakaran khusus dan pengalaman yang relevan mengikut keperluan PTJ.
Syarat Lantikan	Tokoh, perunding, penyelidik, ahli akademik dan pentadbir yang tersohor dalam bidang penyelidikan, pengajaran, atau bidang-bidang tertentu yang berkaitan memberi sumbangan profesional kepada perkembangan sesuatu bidang ilmu
Bayaran	Honorarium sebanyak RM500.00 sehari dan tidak melebihi RM4,000.00 sebulan.
Kemudahan	<u>Faedah Perubatan:</u> Rawatan diri sendiri di Pusat Sejahtera / HPUSM / IPPT / Pusat Kesihatan (Kejuruteraan). <u>Kemudahan Lain:</u> i. Perpustakaan Universiti ii. Penggunaan emel rasmi USM iii. Kad afiliasi (permohonan oleh calon / PTJ) iv. Kemudahan ruang kerja (tertakluk kepada kesesuaian PTJ) Lantikan Bukan Warganegara dengan bayaran, layak menerima faedah berikut sekali dalam satu lantikan: i. Perjalanan Udara – Tambang perjalanan termurah dan terdekat (tertakluk kepada kelulusan & peruntukan Universiti) ii. Penginapan – Dibiayai untuk 8 hari 7 malam (tertakluk kepada kelulusan & peruntukan Universiti)
Peranan dan Sasaran /KPI	Calon yang dilantik perlu memberikan sumbangan dalam sekurang-kurangnya satu (1) Objektif Strategik (SO) yang berikut:

SO	Peranan	Sasaran
SO 1	▪ **Membantu dalam proses pengiktirafan daripada badan-badan profesional bagi program-program dan staf yang berkaitan.** ▪ **Memberi khidmat nasihat dalam pembangunan kurikulum, silibus program pengajian dan pentaksiran pelajar.** ▪ **Menyampaikan syarahan umum / ucaptama dalam bidang kepakaran mengikut keperluan PTJ/USM.**	▪ **Bilangan pengiktirafan.** ▪ **Bilangan bengkel semakan kurikulum yang dihadiri.** ▪ **Bilangan kuliah/syarahan dalam bidang kepakaran mengikut keperluan PTJ/USM.**
SO 2	▪ Membantu meningkatkan kebolehpasaran pelajar. ▪ Membantu keterlibatan pelajar dengan industri/komuniti.	▪ Bilangan program keterlibatan pelajar dengan industri / komuniti.

	SO 3	▪ Membantu dalam aktiviti penyelidikan dan perundingan. ▪ Menyumbang secara kolaboratif dalam penulisan artikel / buku dan lain-lain untuk tujuan penerbitan.	▪ Bilangan aktiviti penyelidikan dan perundingan. ▪ Bilangan artikel / buku dan lain-lain yang diterbitkan.
	SO 4	▪ **Membantu menjalinkan kerjasama antara PTJ dengan industri serta masyarakat.**	▪ **Bilangan jalinan kerjasama.**
	SO 5	▪ **Membantu penjanaan pendapatan PTJ.**	▪ **Bilangan program penjanaan pendapatan PTJ cth: Endowment gifts/ micro-credential/ off-shore/ MOOCs/ short courses.**

KPI yang ditetapkan bagi calon berdasarkan saranan universiti iaitu perlu memberikan sumbangan dalam sekurang-kurangnya satu (1) Objektif Strategik (SO) yang berikut:

SO	Objektif Strategik	Sasaran KPI
SO1	Menyampaikan syarahan umum mengikut keperluan	Bilangan syarahan: 4–8
SO4	Membantu menjalinkan kerjasama antara PTJ dengan industri (*outbound academic staff*) Membantu menjalinkan kerjasama bagi permohonan geran penyelidikan antarabangsa (Jepun)	Bilangan kerjasama (*outbound academic staff*): 4 Bilangan Kerjasama (geran penyelidikan Jepun): 1
SO5	Membantu penjanaan pendapatan PTJ dalam bentuk bengkel/seminar	Bilangan Bengkel Umum: 1

Keputusan:

Jawatankuasa Kecil PTJ – Sumber Manusia (JKPTJSM) Pusat Pengajian Teknologi Industri telah bersidang dan memperakukan palantikan Dr. Quamrul Hassan sebagai Felo Adjung di Pusat Pengajian Teknologi Industri bagi tempoh 1 tahun di bawah peruntukan Universiti.

BIL	CALON	LAMPIRAN
1	Dr. Quamrul Hassan (Bukan Warganegara)	Lampiran G

Mesyuarat ditangguhkan pada jam 12:45 petang dengan ucapan terima kasih daripada Pengerusi kepada semua yang hadir.

Disemak dan disahkan:-

Pengerusi
(PROFESOR MADYA DR. YUSRI YUSUP)

Disediakan oleh:-

Setiausaha
(NOORAIDA MAD NASER)

LAMPIRAN 1A

KEKOSONGAN PERJAWATAN
BIDANG: BIOPROCESS ENGINEERING / BIOTECHNOLOGY ENGINEERING / CHEMICAL ENGINEERING / ENVIRONMENTAL ENGINEERING OR ENGINEERING TECHNOLOGY (APPLIED IN COMPUTER-AIDED DESIGN / SEPARATION AND BIOLOGICAL TREATMENT).

JOB VACANCY

UNIVERSITY LECTURER (CONTRACT – FOR EXPATRIATE)

SCHOOL OF INDUSTRIAL TECHNOLOGY
UNIVERSITI SAINS MALAYSIA

FIELD: BIOPROCESS ENGINEERING / BIOTECHNOLOGY ENGINEERING / CHEMICAL ENGINEERING / ENVIRONMENTAL ENGINEERING OR ENGINEERING TECHNOLOGY (APPLIED IN COMPUTER-AIDED DESIGN / SEPARATION AND BIOLOGICAL TREATMENT).

QUALIFICATION

1. Non-Malaysian citizen.
2. Bachelor's Degree in Bioprocess Engineering / Biotechnology Engineering / Chemical Engineering / Environmental Engineering or Engineering Technology or equivalent from a local or foreign institution of higher learning recognised by the government.
3. Doctor of Philosophy (PhD) degree in a related field of Bioprocess Engineering / Biotechnology Engineering / Chemical Engineering / Environmental Engineering or Engineering Technology or its equivalent from a local or foreign institution of higher learning recognised by the government.

ADDITIONAL CRITERIA

1. Have at least ten (10) publications in Q1/Q2 WoS-indexed with a minimum of six (6) journals as a corresponding or first author or equivalent and demonstrate excellence in research. Candidates who achieved a WoS h-index > 10 will be given priority.
2. Demonstrate creativity and innovation as a dedicated educator, actively involved in teaching, research, consulting, publishing activities and community service.
3. Have good emotional mental health, a positive attitude and is willing to face challenges for university excellence.
4. Candidates will be given preference if they have:
   - Professional certificate (Ir., Ts., Ch.M., etc.) in a related field with two (2) years of teaching experience.
   - Experience in managing, conducting, or networking with international undergraduate or postgraduate student recruitment for school.
   - Experience as a postdoctoral fellow in a recognised institution with active involvement in research and scholarly publications.

TASK SPECIFICATIONS

1. Teach for academic programs at the School of Industrial Technology related to Bioprocess Engineering / Biotechnology Engineering /Chemical Engineering / Environmental Engineering or Engineering Technology (Applied in Computer-Aided Design / Separation / Biological Treatment).
2. Supervise MSc/PhD candidates.
3. Carry out research and publication tasks.
4. Engage in various tasks and services related to the university, community, industry, consulting activities, commercialisation and academic leadership accolades.

HOW TO APPLY:
Kindly forward by email your resume (CV), cover letter and related documents by 26 May 2025 to:

DEAN
School of Industrial Technology
Universiti Sains Malaysia,
11800 PULAU PINANG
Email: dean_ind@usm.my

Note: Only shortlisted applications will be called for the interview session.

LAMPIRAN A

PELANTIKAN BAHARU STAF AKADEMIK (KONTRAK - BUKAN WARGANEGARA / EKSPATRIAT)

BIDANG - BIOPROCESS ENGINEERING / BIOTECHNOLOGY ENGINEERING / CHEMICAL ENGINEERING / ENVIRONMENTAL ENGINEERING OR ENGINEERING TECHNOLOGY (APPLIED IN COMPUTER-AIDED DESIGN / SEPARATION AND BIOLOGICAL TREATMENT).

1-A	**NAMA**	**DR. ZAKIR KHAN (PILIHAN PERTAMA)**
1.	Umur	47 tahun (4 Februari 1978)
2.	Warganegara	Pakistan
3.	Jabatan	Pusat Pengajian Teknologi Industri
4.	Kelayakan Akademik	PhD in Chamical Engineering University Technology PETRONAS, Malaysia 2013 Master in Chamical Engineering Friedrich-Alexander Universität Erlangen-Nürnberg, Germany 2005 BSc in Chemical Engineering University of Engineering and Technology, Peshawar, Pakistan 2001
5.	H-Index 1. Scopus 2. WoS	H-index= 22 (Scopus) H-index= 19 (WoS
6.	Bidang Kepakaran	▪ Biomass/waste-to-energy expert with focus on catalytic gasification, hydrogen production, and carbon capture. ▪ Extensive experience in pilot plant design & operation, and real-time control systems. ▪ Specialized photocatalysis, ORC-based waste heat recovery, and integration with renewable systems. ▪ Process Optimization (via Design Expert Software) ▪ Over 15 years' experience in teaching, supervision, and curriculum development aligned with Washington Accord. ▪ Led/participated in national and international R&D projects (Pakistan Science Foundation, EPSRC (UK), China Academy of Science, PETRONAS, Malaysia). ▪ Head of Biomass Energy Research at COMSATS University Islamabad, Lahore Campus. ▪ Author of 50+ publications (30 Q1), patent holder, and recipient of multiple research awards. ▪ Senior Member – International Association of Carbon Capture; Certified Professional Engineer (PEC).
7.	Pengalaman (Akademik / Industri /dan sebagainya)	**TEACHING EXPERIENCE** Tenured Associate Professor

1-A	NAMA	DR. ZAKIR KHAN (PILIHAN PERTAMA)
		Department of Chemical Engineering, COMSATS University Islamabad, Lahore Campus 05/2023 - to date Assistant Professor Department of Chemical Engineering COMSATS University Islamabad, Lahore Campus 03/2018 – 04/2023 Assistant Professor Department of Chemical Engineering COMSATS Institute of Information Technology, Lahore, Pakistan 07/2013 - 07/2015 Graduate Assistant (Teaching assignments) Department of Chemical Engineering Universiti Teknologi PETRONAS, Malay 02/2009 - 04/2013 Lecturer Department of Chemical Engineering, COMSATS Institute of Information Technology, Lahore, Pakistan 09/2006 - 01/2009 **RESEARCH EXPERIENCE** Head, Biomass Conversion Research Group Department of Chemical Engineering COMSATS University Islamabad, Lahore Campus 05/2023 - to date Post-Doctoral Research Associate (PDRA) School of Engineering, University of Glasgow, UK 08/2015 - 03/2018 Affiliate Staff School of Engineering, University of Glasgow, UK 06/2018 - 02/2020 Research Officer/Graduate Assistant Centre for Biofuel and Biochemical Research, Department of Chemical Engineering Universiti Teknologi PETRONAS, Malaysia 02/2009 - 07/2013 Student Research Assistant Department of Environmental Technology and Recycling, Friedrich-Alexander Universität Erlangen-Nürnberg, German 03/2003 - 03/2004 **INDUSTRIAL EXPERIENCE** Research Assistant Promeos GmbH, Erlangen, Germany 09/2003 - 12/2005

1-A	NAMA	DR. ZAKIR KHAN (PILIHAN PERTAMA)
8.	Senarai Pencapaian Penyelidikan dan Penerbitan (untuk tempoh 5 tahun terkini)	**PUBLICATION/JOURNAL** **2025** 1. Naila Amin, Zakir Khan, Hamad Almohamadi, Muhammad Shahbaz, Sorption-enhanced gasification of municipal solid waste in an updraft gasifier utilizing limestone for dual activity as a catalyst and CO2 adsorbent, Biofuel, Bioproducts and Biorefining, 2025. **2024** 1. Ahmad Naquash, Zeeshan Hameed, Mohammad Abdul Qyyum, Zakir Khan, Ala'a H. Al-Muhtaseb, Amjad Riaz, Moonyong Lee, "Biohydrogen and biomethane production from biomass gasification: compositional analysis, recent advancements, challenges, and prospects", Process Safety and Environmental Protection, 2024 2. Razia Talib, Zakir Khan, Shahzad Khurram, Abrar Inayat, Khurram Shahzad, Ian A. Watson, "Efficiency enhancement of a combined cycle power plant by thermal integration of multiple waste heat streams with organic Rankine cycle, Asia-Pacific Journal of Chemical Engineering, 2024, e.3168. 3. Zakir Khan, Muhammad Shahbaz, Syed Ali Ammar Taqvi, Ahmed AlNouss, Tareq Al-Ansari, Usama Ahmed, "Equilibrium Modelling of Steam Gasification of PKS System and CO2 Sorption Using CaO: A Digitalized Parametric and Techno-Economic Analysis, Journal of Digital Chemical Engineering, 2024, 13, p:100184. 4. Kanwal Rashid, Muhammad Aslam; Ervin Rácz, Sohail Nadeem; Zakir Khan, Nawshad Muhammad, Zeeshan Rashid, Ahmed M. Aljuwayid, Muhammad Kashif Shahid, Muhammad Irfan "Mesoporous silica-grafted deep eutectic solvent-based nanofiltration membranes for water purification: Synthesis, hydrophilicity and heavy metals rejection performance", Nanotechnology Reviews, 2024, 13, pp: 20230213. 5. Zeeshan Hameed, Zakir Khan, Shahzad Khurram, Abrar Inayat, Naila Amin, Muhammad Aslam, Ian Watson, "Co-gasification Study of blends of Municipal Solid Waste with Sugarcane Bagasse and Rice Husk using the Coats-Redfern method, Journal of Energy Institute, 2024; 2024;113, p: 101542. **6.** Naila Amin, Zakir Khan, Abdul Razzaq, Moinuddin Ghauri, Shahzad Khurram, Abrar Inayat, Mujtaba Jaffery, Zeeshan Hameed, "Municipal solid waste air gasification using waste marble powder as a catalyst for syngas production", Journal of Energy Institute, 2024;113, p:101496. **2023** 1. Razia Talib, Zakir Khan, Shahzad Khurram, Abrar Inayat, Moinuddin Ghauri, Mohsin Abbas, Ian Watson, "Energy efficiency enhancement of a thermal power plant by novel heat integration of Internal combustion engine, Boiler, and Organic Rankine Cycle", Asia-Pacific Journal of Chemical Engineering, 2023, p. e3013. **2.** Abrar Inayat, Farrukh Jamil, Shams Forruque Ahmed, Muhammad Ayoube, Peer Mohamed Abdul, Muhammad Aslam, M. Mofijurgi, Zakir Khan, Ahmad Mustafah, "Thermal degradation characteristics, kinetic and thermodynamic analyses of date palm surface fibers at different heating rates", Fuel, 2023, 335 (1), p.127076.

1-A	NAMA	DR. ZAKIR KHAN (PILIHAN PERTAMA)
		2022 1. Zakir Khan, Prashant Kamble, Trevor DiLallo, Willy O'Sullivan, Ellen D. Turner, Andrew Mackay, Paula Blanco-Sanchez, Xi Yu, Anthony Bridgwater, Jon Paul McCalmont, Iain Donnison, Ian Watson, "Design, instrumentation, and operation of a standard downdraft, laboratory-scale gasification testbed utilising novel seed-propagated hybrid Miscanthus pellets" Applied Energy. 2022, 315, p.118864. **2.** Muhammad Saeed, Muhammad Zafar, Abdul Razzaq, Shenawar Ali Khan, Zakir Khan and Woo Young Kim, "Visible-light-active Zn–Fe Layered double hydroxide (LDH) for the photocatalytic conversion of rice husk Extract to value-added products, Applied Sciences, 2022, 12(5), pp: 2313. **2021** 1. Abrar Inayat, Zakir Khan, Muhammad Aslam, Muhammad Shahbaz, Murni M.Ahmad, M.I.Abdul Mutalib, Suzana Yusupe" Heat Integration Modeling of Hydrogen Production from Date Seeds via Steam Gasification" International Journal of Hydrogen Energy, 2021, 46(59), pp: 30592-30605. 2. Abrar Inayat, Zakir Khan, Muhammad Aslam, Muhammad Shahbaz, Murni M.Ahmad, M.I.Abdul Mutalib, Suzana Yusupe" Integrated adsorption steam gasification for enhanced hydrogen production from palm waste at bench scale plant" International Journal of Hydrogen Energy, 2021, 46(59), pp: 30581-30591. **3.** Nada El Darra, Fei Xie, Prashant Kamble, Zakir Khan, Ian Watson, "Decontamination of Escherichia coli on dried onion flakes and black pepper using Infra-red, ultraviolet and ozone hurdle technologies, Heliyon, 2021, 7 (6), p: e072592. **PATENT** **1.** Suzana Yusup, Murni M Ahamd, Zakir Khan, Abrar Inayat, Voi S Chok, Taufiq A Mohammad, "Process of production hydrogen rich gas from biomass". (My-157067-A, PI2012002104 **BOOK** **1.** Suzana Yusup and Zakir Khan. "Enhancement of bioH2 production via in-situ catalytic adsorption (ICA): Optimization and kinetic studies", Lambert Academic Publishing, Germany, 2013 (ISBN: 978-3-659 40782-6). **BOOK CHAPTERS** 1. Abrar Inayat, Rumaisa Tariq, Chaouki Ghenai, Muhammad Shahbaz, Zakir Khan, Farrukh Jamil, Muhammad Ayoub. "Recent progress in biomass air gasification using moving and fixed bed gasifiers". Innovations in Thermochemical Technologies for Biofuel Processing, Chapter 10, pp. 237-258 (ISBN: 978-0-323-85586-0) Elsevier, Amsterdam, Netherlands, (2022) 2. Abrar Inayat, Rumaisa Tariq, Ola Alsaidi, Muhammad Shahbaz, Zakir Khan, Chaouki Ghenai and Tareq Al-Ansari. "Recent progress in modeling and simulation of biomass conversion to Biohydrogen", Value-Chain of Biofuels (978-0-12-824388-6), Chapter 13, Elsevier, Amsterdam, Netherlands, (2021).

1-A	NAMA	DR. ZAKIR KHAN (PILIHAN PERTAMA)
		3. Mehmood Ali, Muhammad Saleem, Zakir Khan and Ian Watson, "The use of crop residues for biofuel production", in Biomass, Biopolymer Based Materials, and Bioenergy: Construction, Biomedical, and other Industrial Applications (978-0-08-102427-0), Woodhead Publishing, Elsevier, Kidlington, UK, 2019. **RESEARCH FUNDINGS/PROJECTS** 1. Development and operation of lab/pilot scale facilities for co-utilization of indigenous low rank coal, biomass, sewage sludge for cleaner energy production for fertilizer industry Role: Investigator/Team Member (PI from NUST Islamabad, Pakistan) Funding body: Pakistan Science Foundation (PSF) and Ministry of Science & Technology (MoST) Funding amount: 6.745 million Pak RS Duration: 36 months (02/2021 - 12/2024) 2. Basic research and capacity investigation for distributed bioenergy utilization via thermochemical conversion Role: Principal Investigator Funding body: Pakistan Science Foundation (PSF) and Chinese Academy of Sciences (CAS), China Funding amount: Pak RS 8.56 million Duration: 48 months (extended for one year) (04/2018 – 12/2021) **PROFESSIONAL DEVELOPMENT (TRAINING/WORKSHOP)** **1.** Korea International Cooperation Agency (KOICA)'s Fellowship on "Training Program on Renewable Energy Technologies (Pakistan)", 2021. **CONFERENCE/WORKSHOPS ORGANIZATION** **1.** Organized Summer Camp, "1st Chemical Engineering and Analytical Techniques Summer Camp" COMSATS University Islamabad, Lahore Campus, Pakistan, 29 July – 03 August 2019.
9.	Senarai Penglibatan Pengajaran dan Penyeliaan (untuk tempoh 5 tahun terkini)	**Teaching** **COMSATS University Islamabad, Lahore Campus, Pakistan** **<u>Undergraduate</u>** • Fluid Mechanics for Chemical Engineers (Dual Degree Program with Lancaster University, UK) (teaching evaluation, 4.6/5.0) • Fuels and Energy (teaching evaluation, 4.2/5.0). • Coal Utilization Technologies (not available) **<u>Postgraduate</u>** • Waste to Energy (teaching evaluation, 4.8/5.0) • Solid Waste Engineering (teaching evaluation, 5.0/5.0) • Combustion Engineering (teaching evaluation, 5.0/5.0) • Fluidization Engineering (teaching evaluation, 5.0/5.0) **Supervision** **COMSATS University Islamabad, Lahore Campus, Pakistan** **<u>Postgraduate</u>**

1-A	NAMA	DR. ZAKIR KHAN (PILIHAN PERTAMA)
		• Naila Amin (PhD, 2025), Catalytic gasification of municipal solid waste for energy production, Supervisor (Completed). • Zeeshan Hameed (PhD, 2025), Gasification of municipal solid waste and biomass as a blend in a fixed bed gasifier for energy production, Supervisor (ongoing). • Razia Talib (PhD, 2025), Cycle efficiency enhancement of combined cycle powerplant by optimal utilization of multiple waste heat streams using Organic Rankin cycle, Co-Supervisor (ongoing). • Muhammad Saeed (MS, 2021), Zn based LDH for photocatalytic biomass conversion to useful/value added chemicals, Supervisor. • Shafqat Abbas (MS, 2022) Biomass gasification for small scale power generation, Mentored/Trained. **<u>Undergraduate</u> (Final Year Design Project)** • Muhammad Abdullah (2024), Transforming Biomethane into green H2: A leap towards a greener Pakistan. (Won prestigious Pakistan Engineering Council financial award among 102 Universities) • Sahar Khan (2023), Production of aviation (jet) fuel via Fischer-Tropsch process (FTP) utilizing biomass gasification. (Selected to be showcased in the 5th Engineering CAPSTONE Expo-2024, Pakistan Engineering Council) • Muhammad Shahid (2022), 10MW power generation in a circulating fluidized bed gasification plant utilizing municipal solid waste. • Shahbaz Naeem (2021), Flue-gas emissions control from the combustion of 5500 tonnes/day coal containing high sulphur and nitrogen contents. • M. Shahbaz Naeem (2019), A plant design project on 300 ton/day Gasification of solid MSW using Fluidized bed
10.	Senarai Pencapaian Sanjungan Akademik dan Anugerah (untuk tempoh 5 tahun terkini)	**Awards and Career Achievements** 1. <u>Head, Biomass Conversion Research Group</u> Department of Chemical Engineering, COMSATS University Islamabad, Lahore Campus, May 2023. 2. <u>Departmental Appreciation Award</u> for exceptional supervision of outstanding Final Year Design Projects (FYDPs), June 2024. **3.** <u>Secured Pakistan Engineering Council Financing</u> out of 102 participating universities for the UG Final Year Design Project titled "Transforming Biomethane into Green Hydrogen: A Leap Towards a Greener Pakistan" at the 5th Engineering Capstone Expo 2024.
11.	Senarai Penglibatan Konsultansi (untuk tempoh 5 tahun terkini)	**Projects Consultations** 1. Industrial client based in Italy: A joint project consultancy with the University of Sharjah (2025), Development of H2 generation system using coffee wastes as feedstock. 2. University of Burgundy, France: Hydrogen generation and purification utilizing biomass in fluidized bed reactor.

1-A	NAMA	DR. ZAKIR KHAN (PILIHAN PERTAMA)
		3. Dr Moinuddin Ghauri, COMSATS University Islamabad, Lahore Campus (2020), Development/modifications to updraft gasifier for production of synthesis gas from biomass wastes.
12.	Senarai Penglibatan Sumbangan Kepada Universiti dan Komuniti (untuk tempoh 5 tahun terkini)	• Program Coordinator, Postgraduate Studies • Campus Manager, Environmental Health & Safety • Spokesman, Afghan Student (Campus) • Convener, Departmental Sport Committee • Convener, Sustainable Future Programs Committee • Convener, Inquiry Committee of Missing Multimedia • Convener, UG Lab Committee • Member/Secretary, Departmental Advisory Committee • Member/Secretary, Departmental Academic Review Committee • Member, Campus Purchase Committee • Member, Department Admission Committee • Member, Board of Study (As a Senior Assistant Professor)
13.	Maklumat Tambahan Calon (Sekiranya Ada)	**PROFESSIONAL MEMBERSHIP** 1. Senior Member, International Association of Carbon Capture (IACC), Since May 2023 2. Approved PhD Supervisor, Higher Education Commission (HEC), Since 2015 3. Lifetime -Certified Professional Engineer, Pakistan Engineering Council, Since 2002 **PROFESSIONAL ROLE & MANAGEMENT** 1. At CUI Lahore Campus (2023-to date), Head of the “Biomass Conversion Research Group”, development of lab and pilot scale updraft gasification systems by exploiting local agricultural wastes and MSW. 2. As a sole PDRA & Affiliate Staff (Aug 2015-Feb 2020), Led and coordinated the project “Real time control of gasifiers to increase tolerance to biomass variety and reduce emissions” at the University of Glasgow, UK. All project-related administrative work within the UK project partners Universities of Glasgow, Aston, Manchester, and Aberystwyth. At the University of Glasgow, led a group of researchers (BEng, MEng and PhD) to smoothly run the different goals of the project. 3. Led and managed, the “Student Proctorial Team” in the Department of Chemical Engineering, COMSATS Institute of Information Technology, Lahore, 2014. 4. Managed the poster exhibition as Team Head in Energy Systems for Sustainable Development (ESSD), Lahore, Pakistan, 20-22 May 2015. 5. At Universiti Teknologi PETRONAS, Malaysia (2009-2013), Led and managed the development of a pilot scale gasification system, and other research groups related to feed pre-treatment/characterization and CO2 adsorption material under the project. 6. Promeos GmbH, Erlangen, Germany (2004), Lead researcher, investigation of thermo-acoustic noise emitted by gas burners.

1-A	NAMA	DR. ZAKIR KHAN (PILIHAN PERTAMA)
		7. Promeos GmbH, Erlangen, Germany (2004), Lead researcher, characterizing different porous medium burner configurations using Infrared (IR) camera. 8. **PROFESSIONAL ROLE AS A TEAM MEMBER** 1. Member Board of Study (BOS) to develop/change curriculum at undergraduate and postgraduate levels. 2. Member/Secretary Departmental Advisory Committee to seek PG activities 3. Member of the Departmental Academic Review Committee to look after departmental affairs. Member of Departmental Coordination and Steering Committee to seek excellence in engineering and research.
14.	Maklumat Sasaran KPI Untuk Tujuan Pelantikan	**1. M1: Penerbitan / Publications** a. Minimum empat (4) penerbitan setahun - penerbitan dalam jurnal berindeks sebagai *First Author/Corresponding Author* *A minimum of four (4) publications per year - publications in indexed journals as the First Author/ Corresponding Author* **2. M1: Penyelidikan / Research** a. 1 geran jangka pendek / kebangsaan/industri/antarabangsa setahun *1 short term grant / national/industrial/international grant per year* **3. M2: Pengajaran / *Teaching*** a. Minimum 8 unit per sidang akademik *Minimum 8 units per academic session* **4. M2: Penyeliaan / Supervision** a. Penyelian 3 - 5 pelajar FYP dan 3 - 5 Latihan Industri per sidang *Supervision: 3-5 FYP students and 3-5 Industrial Training students per session* b. 1 pelajar pasca siswazah (Sarjana) / *1 postgraduate student (Master's)* c. 1 pelajar pasca siswazah (PhD) *1 postgraduate student (PhD)* **5. Penjanaan Pendapatan / Income Generation** a. Minimum 2 aktiviti penjanaan kewangan seperti konsultasi, kursus-kursus profesional dan lain-lain b. *Minimum of 2 financial generation activities such as consultations, professional courses, etc.* Prestasi staf akan dinilai setiap tahun bagi memastikan sasaran KPI yang telah ditetapkan tercapai. *The performance of staff will be reviewed yearly to ensure the fulfillment of the designated KPI targets.*

1-A	NAMA	DR. ZAKIR KHAN (PILIHAN PERTAMA)
15.	Penetapan Gaji	Gaji Semasa Gred DS51 : RM6,210.00 Imbuhan Tetap Keraian : RM600.00 Imbuhan Tetap Perumahan : RM700.00 Bayaran Bantuan Sara Hidup : RM300.00 **Jumlah Keseluruhan : RM7,810.00**
16.	Justifikasi Lantikan	1. Pelantikan ini adalah untuk memenuhi kekurangan tenaga pengajar dan memenuhi kekosongan bagi menggantikan satu (1) orang Pensyarah Universiti (Kontrak – Bukan Warganegara/Ekspatriat) di Bahagian Teknologi yang tamat perkhidmatan: ▪ Dr. Esam Bashir Abdulsalam Yahya (01.09.2022 - 23.04.2025) 2. Keperluan kepakaran dalam pembelajarafsumarnin dan pengajaran. Bertanggungjawab dalam semua aspek yang berhubungan dengan pelaksanaan tugas pengajaran dan pembelajaran di Pusat Pengajian Teknologi Industri seperti yang diperlukan dan ditetapkan oleh PTJ. 3. Keperluan kepakaran dalam penyelidikan dan penyeliaan pelajar ijazah tinggi (sumbangan kepada penyelidikan, penyeliaan dan penerbitan) yang dapat membantu meningkatkan prestasi Bahagian, Pusat Pengajian dan Universiti. 4. Pelantikan ini dapat menyumbang kepakaran dalam meningkatkan prestasi dan KPI pusat pengajian. 5. Pelantikan ini akan dapat memastikan perancangan dan hala tuju pusat pengajian tidak terjejas disebabkan kekurangan tenaga pengajar dalam bidang ini. 6. Memenuhi bidang tujahan yang diperlukan.
17.	Perakuan Jawatankuasa Kecil PTJ	Setelah menemuduga calon, Jawatankuasa Kecil – Sumber Manusia berpuas hati dan bersetuju memperakukan pelantikan Dr. Zakir Khan sebagai Pensyarah Universiti DS13 (Kontrak – Bukan Warganegara) kerana memenuhi semua syarat dan keutamaan yang diperlukan serta justifikasi berikut: 1. Pelantikan ini adalah untuk memenuhi kekosongan jawatan di Bahagian Teknologi Bioproses, Pusat Pengajian Teknologi Industri. 2. Memenuhi bidang tujahan yang diperlukan. 3. Rekod penerbitan yang sangat baik. 4. Berpotensi untuk melonjak KPI PTJ. 5. Calon mempunyai pengalaman dalam pengajaran dan pembelajaran. Pelantikan ini bagi tempoh dua (2) tahun berkuat kuasa pada tarikh melapor diri.

1-A	NAMA	DR. ZAKIR KHAN (PILIHAN PERTAMA)
18.	Catatan Tambahan PTJ	**<u>Tugasan Pengajaran dan Pembelajaran (P&P)</u>** Antara kursus yang terlibat: 1. IEK 212/3 Process Heat Transfer 2. IEK 213/3 Mass Transfer and Separation Process 3. IEA 216/3 Computer Application in Industry 4. IBK 314 Downstream Process Technology 5. IBA407 Practical in Downstream Processing 6. IBG307 Bioprocess Optimization And Simulation
19.	Perakuan Jawatankuasa Pemilih	-
20.	Nota Rujukan JKPTJSM	

<table>
<tr><th>2-B</th><th>NAMA</th><th>DR. TS. IRVAN DAHLAN MANSYOER (PILIHAN KE DUA)</th></tr>
<tr><td>1.</td><td>Umur</td><td>51 tahun (8 February 1974)</td></tr>
<tr><td>2.</td><td>Warganegara</td><td>Indonesia (Malaysian permanent resident)</td></tr>
<tr><td>3.</td><td>Jabatan</td><td>Pusat Pengajian Teknologi Industri</td></tr>
<tr><td>4.</td><td>Kelayakan Akademik</td><td>PhD in Chemical Engineering
Universiti Sains Malaysia, MALAYSIA
2009

M.Sc. in Chemical Engineering
Universiti Sains Malaysia, MALAYSIA
2004

Bachelor of Engineering (Chemical Engineering)
Universitas Syiah Kuala (USK), Aceh, INDONESIA
1999</td></tr>
<tr><td>5.</td><td>H-Index
1. Scopus
2. WoS</td><td><table>
<tr><td>Profiles*</td><td>ResearcherID</td><td>Scopus</td><td>Google Scholar</td></tr>
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<tr><td>Citations</td><td>868</td><td>1124</td><td>1815</td></tr>
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<tr><td>6.</td><td>Bidang Kepakaran</td><td>Adsorption, Wastewater & Air pollution treatment</td></tr>
<tr><td>7.</td><td>Bidang Lantikan</td><td>Teknologi Bioproses / Teknologi Makanan</td></tr>
<tr><td>8.</td><td>Pengalaman (Akademik / Industri /dan sebagainya)</td><td>WORKING EXPERIENCE

Present Position:
Senior Lecturer
School of Chemical Engineering, Universiti Sains Malaysia
13 August 2012 – 31 December 2025

Senior Lecturer
Universiti Sains Malaysia (School of Civil Engineering)
13 August 2009 – 12 August 2012

Graduate Assistant (Teaching)
Universiti Sains Malaysia (School of Chemical Engineering)
July 2005 - April 2008

Graduate Research Assistant
Universiti Sains Malaysia (School of Chemical Engineering)
March 2003 - June 2005</td></tr>
<tr><td>9.</td><td>Senarai Pencapaian Penyelidikan dan Penerbitan (untuk tempoh 5 tahun terkini)</td><td>PUBLICATION/JOURNAL

Refereed Journals</td></tr>
</table>

2-B	NAMA	DR. TS. IRVAN DAHLAN MANSYOER (PILIHAN KE DUA)
		2024 1. Dahlan, I.* & Suhaimi, M.H.M. (2025). Adsorption of CO_2 using NaOH-modified nanoclay montmorillonite adsorbent: Comparative analysis of RSM-based central composite design and ANN-based models in modelling and optimization. ***Arabian Journal for Science and Engineering***. Galley Proof. https://doi.org/10.1007/s13369-025-10401-9. (2024 IF: 2.9, SCIE/Scopus, JCR: Q2, SJR: Q1). 2. Shafti, D.M., Dahlan, I.* & Din, A.T.M. (2025). Removal of acid violet 7 (AV7) dye from aqueous solutions using modified MOF-5 adsorbents. ***Water, Air, and Soil Pollution***, 236, 63. https://doi.org/10.1007/s11270-024-07708-y. (2024 IF: 3.0, SCIE/Scopus, JCR: Q2, SJR: Q2). 3. Dahlan, I.*, Obi, C.C. & Poopathi, V. (2025). Predicting the adsorption capacity of organic dye using zirconium-based metal–organic framework: A comparative analysis of RSM and ANN-based models. ***Biomass Conversion and Biorefinery*** 15:285–295. https://doi.org/10.1007/s13399-023-05012-z. (2024 IF: 4.1, SCIE /Scopus, JCR: Q2, SJR: Q2). 4. Dahlan, I.*, Yee, M.K., Shafawi, A.N. & Amer, N.M. (2025). Siliceous waste material supported MOF-5 for carbon monoxide capture at low temperature. ***Biomass Conversion and Biorefinery*** 15: 6195–6206. https://doi.org/10.1007/s13399-024-05564-8. (2024 IF: 4.1, SCIE /Scopus, JCR: Q2, SJR: Q2). 5. Hasan Harahsheh, Y.A., Dahlan, I.* & Abdullah, A.Z. (2024). Adsorption of malachite green dyes from aqueous solution by metal organic framework-5 incorporated coal fly ash (MOF-5/CFA). ***Journal of Dispersion Science and Technology***, Online, https://doi.org/10.1080/01932691.2024.2425968. (2024 IF: 1.9, SCIE /Scopus, JCR: Q4, SJR: Q3). 6. Shafti, D.M., Dahlan, I.* & Din, A.T.M. (2024). A review of the effectiveness of metal-organic frameworks in removing dye effluents. ***Water Practice and Technology***, 19 (12): 4699–4733. https://doi.org/10.2166/wpt.2024.279. (2024 IF: 1.8, ESCI/Scopus, JCR: Q3, SJR: Q3). 7. Dahlan, I.*, Aziz, H.A. & Hung, Y-T. (2024). Recent development of metal-organic framework 5 adsorbents for organic dye removal from aqueous solution. ***International Journal of Environment and Waste Management***, 35(3): 378 – 389. https://doi.org/10.1504/IJEWM.2024.142506. (2024 IF:0.5, ESCI/Scopus, JCR: Q4, SJR: Q4). 8. Dahlan, I.*, Obi, C.C., Razaman, N.S. & Hasan, H.Y.A. (2024). Adsorptive decolorization of brilliant green dye in aqueous media using various modified MOF-5 adsorbents. ***Groundwater for Sustainable Development*** 26, 101228. https://doi.org/10.1016/j.gsd.2024.101228. (2024 IF: 5.6, ESCI/Scopus, JCR: Q1, SJR: Q1). 9. Mulkan, A., Zulkifli, N.W.M., Husin, H., Ahmadi, & Dahlan, I. (2024). Performance and emissions assessment under full load operation of an unmodified diesel engine running on biodiesel-based waste cooking oil synthesized using JPW solid catalyst. ***Renewable Energy***, 224: 120145. https://doi.org/10.1016/j.renene.2024.120145. (2024 IF: 9.1, SCIE /Scopus, JCR: Q1, SJR: Q1). **2023**

2-B	NAMA	DR. TS. IRVAN DAHLAN MANSYOER (PILIHAN KE DUA)
		1. Mulkan, A., Zulkifli, N.W.M., Husin, H., Ahmadi, Dahlan, I. & Syafiie, S. (2023). Development of jackfruit (*Artocarpus heterophyllus*) peel waste as a new solid catalyst: Biodiesel synthesis, optimization and characterization. ***Process Safety and Environmental Protection***, 177: 152–168. https://doi.org/10.1016/j.psep.2023.07.021. (2023 IF: 6.9, SCIE /Scopus, JCR: Q1, SJR: Q1). 2. Iheanacho, O.C., Nwabanne, J.T., Obi, C.C., Igwegbe, C.A., Onu, C.E. & Dahlan, I. (2023). Adsorptive dephenolization of aqueous solutions using thermally modified corn cob: Mechanisms, point of zero charge, and isosteric heats studies. ***Adsorption Science & Technology***, Volume 2023, Article ID 2813663. https://doi.org/10.1155/2023/2813663. (2023 IF: 2.8, SCIE /Scopus, JCR: Q2, SJR: Q2). 3. Dahlan, I.*, Keat, O.H., Aziz, H.A. & Hung, Y-T. (2023). Synthesis and characterization of MOF-5 incorporated waste-derived siliceous materials for the removal of malachite green dye from aqueous solution. ***Sustainable Chemistry and Pharmacy***, 31:100954. https://doi.org/10.1016/j.scp.2022.100954. (2023 IF: 5.5, SCIE/Scopus, JCR: Q1, SJR: Q1). **2022** 1. Dahlan, I.*, Mazlan, W.H.W., Mulkan, A., Zwain, H.M., Hassan, S.R., Aziz, H.A., Hasan, H.Y.A. & Zekker, I. (2022). Modeling of batch organic dye adsorption using modified metal-organic framework-5. ***Chemical Engineering & Technology***, 45(11): 2080-2087. https://doi.org/10.1002/ceat.202200042. (2022 IF: 2.1, SCIE/Scopus, JCR: Q3, SJR: Q2). 2. Dahlan, I.*, Azhar, E.E.M., Hassan, S.R., Aziz, A.A. & Hung, Y-T. (2022). Statistical modeling and optimization of process parameters for 2,4-dichlorophenoxyacetic acid removal by using AC/PDMAEMA hydrogel adsorbent: Comparison of different RSM designs and ANN training methods. ***Water*** 14(19), 3061. https://doi.org/10.3390/w14193061. (2022 IF: 3.4, SCIE/Scopus, JCR: Q2, SJR: Q1). 3. Zwain, H.M., Barghash, H., Vakili, M. Majdi, H. Sh. & Dahlan, I.* (2022). Modeling and optimization of process parametric interaction during high-rate anaerobic digestion of recycled paper mill wastewater using the response surface methodology. ***Water Reuse*** 12(1): 78-91. https://doi.org/10.2166/wrd.2022.088. (2022 IF: 4.5, SCIE/Scopus, JCR: Q2, SJR: Q2). 4. Hassan, S.R., Hung, Y-T., Dahlan, I.* & Aziz, H.A. (2022). Kinetic study of the anaerobic digestion of recycled paper mill effluent (RPME) by using a novel modified anaerobic hybrid baffled (MAHB) reactor. Water 14(3), 390. https://doi.org/10.3390/w14030390. (2022 IF: 3.4, SCIE/Scopus, JCR: Q2, SJR: Q1). **2021** 1. Dahlan, I.*, Hassan, S.R. & Lee, W.J. (2021). Modeling of modified anaerobic baffled reactor for recycled paper mill effluent treatment using response surface methodology and artificial neural network. ***Separation Science and Technology***, 56(3): 592-603. https://doi.org/10.1080/01496395.2020.1728321. (2021 IF: 2.799, SCIE/Scopus, JCR: Q3, SJR: Q2).

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		2. Zwain, H.M., Al-Marzook, F.A., Nile, B.K., Jeddoa, Z.M.A., Atallah, A.H., Dahlan, I.*, & Hassan, W.H. (2021). Morphology analysis and microbial diversity in novel anaerobic baffled reactor treating recycled paper mill wastewater. ***Archives of Environmental Protection***, 47(4): 9–17. https://doi.org/10.24425/aep.2021.139498. (2021 IF: 1.872, SCIE/Scopus, JCR: Q4, SJR: Q2). 3. Zwain, H.M., Naje, A.S., Vakili, M. & Dahlan, I.* (2021). Temperature analysis of a novel MAIB reactor during the treatment of wastewater from recycled paper mill. ***Water Practice and Technology***, 16(2): 592-604. https://doi.org/10.2166/wpt.2021.023. (2021 IF: n/a, ESCI/Scopus, JCR: Q4, SJR: Q3). 4. Zwain, H.M., Almurshedi, K.R., Vakili, M., Dahlan, I. & Naje, A.S. (2021). Water quality and radionuclides content assessment of Al-Najaf sea: Case study. ***Journal of Ecological Engineering***, 22(2): 262-271. https://doi.org/10.12911/22998993/131183. (2021 IF: n/a, ESCI/Scopus, JCR: Q4, SJR: Q3). 5. Dahlan, I.* & Ling, N.W. (2021). Adsorption of acid violet 7 (AV7) dye using RHA-CFA adsorbent: Modelling, process analysis and optimization. ***Separation Science and Technology***, 56(1): 54-67. https://doi.org/10.1080/01496395.2019.1708115. (2021 IF: 2.799, SCIE/Scopus, JCR: Q3, SJR: Q2). **2020** 1. Zwain, H.M., Nile, B.K., Faris, A.M., Vakili, M. & Dahlan, I. (2020). Modelling of hydrogen sulfide fate and emissions in extended aeration sewage treatment plant using Toxchem simulations. ***Scientific Reports***, 10, 22209. https://doi.org/10.1038/s41598-020-79395-8. (2020 IF: 4.379, SCIE/Scopus, JCR: Q1, SJR: Q1). 2. Hassan, S.R. & Dahlan, I.* (2020). Investigation and optimization of a novel MAHB reactor for COD and lignin removal and methane production using response surface methodology (RSM) and artificial neural network (ANN). Bulgarian Chemical Communications, 52(1):102-113. http://bcc.bas.bg/BCC_Volumes/Volume_52_Number_1_2020/BCC-52-1-2020-102-113-Hassan-5184.pdf. (2020 IF: n/a, SCIE/Scopus, JCR: n/a, SJR: Q4). 3. Hassan, S.R., Zaman, N.Q. & Dahlan, I.* (2020). Evaluation of recycled paper mill effluent digestion in a modified anaerobic hybrid baffled (MAHB) reactor: Reactor performance & kinetic studies. ***Preparative Biochemistry & Biotechnology***, 50(3): 234–239. https://doi.org/10.1080/10826068.2019.1692214. (2020 IF: 2.162, SCIE/Scopus, JCR: Q3, SJR: Q3). **BOOK/BOOK CHAPTER** 1. Dahlan, I.*, Keat, O.H, Mulkan, A., Aziz, H.A. & Hung, Y-T. (2024). Metal-organic framework 5 (MOF-5) composites: From synthesis optimization to its application in textile industry wastewater treatment. In: Handbook of Industrial and Hazardous Wastes Treatment (Editors: Lawrence K. Wang, Mu-Hao S. Wang, & Yung-Tse Hung), CRC Press. Under review. 2. Dahlan, I. *, Mahdzir, S., Mulkan, A. & Zwain, H.M. (2022). Comparative adsorption study of acid violet 7 and brilliant green dyes in aqueous media using rice husk ash (RHA) and coal fly ash (CFA)

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		mixture (Chapter 7). https://doi.org/10.2174/9789815039924122010009. In: Sustainability Studies: Environmental and Energy Management (Editors: G. Venkatesan, S.L. Prabu & M. Rengasamy), Bentham Science Publishers Pte.Ltd., Singapore, pp. 129-147. ISBN: 978- 981-5039-93-1 (Print), ISBN: 978-981-5039-92-4 (Online). https://doi.org/10.2174/9789815039924122010 1. 3. Dahlan, I.*, Yusoff, M.S., Akinbile, C.O., Wang, L.K. & Wang, M-H. S. (2022). Restaurant waste recycle and disposal. In: Solid Waste Engineering and Management, 8 Vol. 2 (Editors: L. K. Wang, M-H. S. Wang & Y-T. Hung), Handbook of Environmental Engineering (HEE, Vol. 24), Springer, pp. 439-542. https://doi.org/10.1007/978-3-030-89336-1_7. **4.** Farraji, H., Dahlan, I. & Eslamian, S. (2021). Water recycling from palm oil mill effluent (Chapter 20). In: Handbook of Water Harvesting and Conservation: Basic Concepts and Fundamentals (Editors: S. Eslamian and F. Eslamian), Wiley-Blackwell, Hoboken, New Jersey, pp. 307-320. Print ISBN: 9781119478959. Online ISBN: 9781119478911. DOI:10.1002/9781119478911 **CONFERENCE** 1. Shafti, D.M., Dahlan, I.*, Din, A.T.M. (2024). Comparative isotherm and kinetic analysis of acid violet 7 (AV7) and methylene blue (MB) adsorption onto MOF-5. In Abstract Book of the 6th International Conference on Applied Engineering and Natural Sciences (ICAENS2024). Paper No. Y-75, p. 102, September 25-26, Virtual Conference, Konya, Turkey. 2. Shafti, D.M., Dahlan, I.*, Din, A.T.M. (2024). Comparative removal of acid violet (AV7) and malachite green (MG) dyes using MOF-5 adsorbents. In Proceeding Book of the 5th International Conference on Engineering and Applied Natural Sciences (ICEANS2024). Paper No. Y-207, p. 543 – 554, August 25-26, Virtual Conference, Konya, Turkey. 3. Dahlan, I.*, Wan Mazlan, W.H., Mulkan, A. & Zwain, H.M. (2023). Modelling of batch organic dye adsorption using modified metal-organic framework-5 (MOF-5). AIP Conference Proceedings 2785, 030024 (2023). (Indexed by Web of Science, Scopus). DOI: https://doi.org/10.1063/5.0147966. 4. Gonawan, F.N., Dahlan, I. & Kamaruddin, A.H. (2023). Physicochemical properties comparison of deep eutectic mixtures of choline chloride with urea, glucose and citric acid. AIP Conference Proceedings 2785, 030018 (2023). (Indexed by Web of Science, Scopus). https://doi.org/10.1063/5.0147979. 5. Dahlan, I.* (2021). Towards the use of metal-organic framework MOF-5 for the removal of organic dye from aqueous solution: Recent research review. In Proceeding Book of the 1st International Conference on Applied Engineering and Natural Sciences (ICAENS 2021) (Paper_351, p. 317), November 1 – 3, Virtual Conference, Konya, Turkey. 6. Mulkan, A. & Dahlan, I. (2021). In-cylinder pressure analysis and performance of IC engine running on producer gas from downdraft biomass gasifier. In Proceeding Book of the 1st International

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		Conference on Applied Engineering and Natural Sciences (ICAENS 2021). Paper_274, p. 264, November 1 – 3, Virtual Conference, Konya, Turkey. 7. Mulkan, A., Zulfadli, T. & Dahlan, I. (2021). Study on preliminary feasibility of solarwind energy to power up water pump for irrigation in the village: Case study in Aceh Besar district, Aceh Province, Indonesia. In Proceedings (The Book of Full Texts) of the 3. International Baku Scientific Research Congress (pp. 377-384), October 15-16, Virtual Conference, Baku, Azerbaijan. 8. Dahlan, I.*, Wan Mazlan, W.H., Mulkan, A. & Zwain, H.M. (2021). Modelling of batch organic dye adsorption using modified metal-organic framework-5 (MOF-5). In e-Book (Abstract Book) of the ICENV−RCEnvE2021 (14th AUN/SEED-Net Regional Conference on Environmental Engineering 2021, in conjunction with 7th International Conference on Environment 2021), (Paper ID. ICENV2021-0078, p. 77), October 6-7, Virtual Conference, Universiti Sains Malaysia, Malaysia. 9. Gonawan, F.N., Dahlan, I. & Kamaruddin, A.H. (2021). Physicochemical properties comparison of deep eutectic mixtures of choline chloride with urea, glucose and citric acid. In e-Book (Abstract Book) of the ICENV−RCEnvE2021 (14th AUN/SEED-Net Regional Conference on Environmental Engineering 2021, in conjunction with 7th 17 International Conference on Environment 2021), (Paper ID. ICENV2021-0066, p. 69), October 6-7, Virtual Conference, Universiti Sains Malaysia, Malaysia. 10. Dahlan, I.*, Ali, N.W. & Mulkan, A. (2021). Magnetic RHA-Fe(III) hydrogels as catalysts for H2 generation from hydrolysis of ammonia borane (NH3BH3). In Proceedings Book of the Hagia Sophia 3. International Conference on Multidisciplinary Scientific Studies (pp. 547–554), September 15-16, Virtual Conference, Istanbul, Turkey. **11.** Dahlan, I.*, Anuar, M.H.A., Zwain, H.M., Hassan, S.R. & Mulkan, A. (2021). Physical and chemical characteristics of microorganisms developed in a modified anaerobic inclining-baffled reactor treating recycled paper mill effluent. In Proceedings Book of the UMTEB International Congress on Vocational & Technical Sciences-XI (pp. 334– 342), September 3-5, Virtual Conference, Ankara, Turkey. **GRANTS** **Principal Investigator** 1. **Title:** Integrated Pyrolysis System for Sustainable Upcycling of Plastic Waste (Project: Removal of pyrolysis multi-pollutant gases using modified metal-organic framework-5) **Sponsor (duration):** Research University (RU) – Team Grant, Universiti Sains Malaysia (2022 – 2024), extended to June 2025. **Members:** Dr. Pooya Lahijani Amiri **Amount:** RM 66,220.00 (Total: RM 325,000.00)
10.	Senarai Penglibatan Pengajaran dan Penyeliaan (untuk tempoh 5 tahun terkini)	**(1) List of courses taught (undergraduate)** EUM 113/3 – Engineering Calculus (Tutor) EUM 114/3 – Advanced Engineering Calculus (Tutor) EMM 101/3 – Engineering Mechanics (Tutor)

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		EKC 113/3 – Physical Chemistry EKC 157/2 – Chemical Engineering Drawing EKC 291/2 – Chemical Engineering Laboratory I SHE 101/2 – Ethnic Relations (Project supervisor) HFE 224/2 – Appreciation of Ethics and Civilizations (Project supervisor) EKC 367/3 – Plant Safety EKC 451/4 – Process Design and Analysis EKC 455/4 – Plant Design and Operation EKC 475/3 – Wastewater Treatment Engineering EKC 396/2 – Chemical Engineering Laboratory III EKC 499/6 – Final Year Project (FYP supervisor) (**2) FYP supervision (undergraduate)** 14 completed & 3 on-going **(3) Postgraduate supervision** 2 MSc (viva completed) & 1 PhD (on-going)
11.	Senarai Pencapaian Sanjungan Akademik dan Anugerah (untuk tempoh 5 tahun terkini)	**Honors and Awards** 1. Award for Hadiah Sanjungan 2024 (Merit Reward), Publication Category for the year 2023, Universiti Sains Malaysia. 2. Award for Hadiah Sanjungan 2023 (Merit Reward), Publication Category for the year 2022, Universiti Sains Malaysia. **3.** NISA Award for Outstanding Iraqi Research 2020 (Best Collaborative Paper), The Network of Iraqi Scientists Abroad (NISA), Iraq. **Other Achievements** Workshop / Short Course / Training Attended various workshops/seminar related to CQI course outcome, Effective teaching for outcomes-based higher education, Blended learning, Research grant & scientific writing, Writing course outcome, USM E-learning training, etc.
12.	Senarai Penglibatan Konsultansi (untuk tempoh 5 tahun terkini)	-
13.	Senarai Penglibatan Sumbangan Kepada Universiti dan Komuniti (untuk tempoh 5 tahun terkini)	1. Technical Program Committee, in 2nd Asia Conference on Renewable Energy and Environmental Engineering (AREEE 2021), Online, March 27-29, 2021, Singapore. 2. Technical Committee, in 6th GoGreen Summit & Awareness on Climate Change, July 1-2, 2020, Hotel Santika Premiere Malang, Malang, Indonesia. 3. Actively involved as a reviewer for high-impact international journals and conferences
14.	Maklumat Tambahan Calon (Sekiranya Ada)	**EDITORIAL/MEMBERSHIP/PROFESSIONAL/ACADEMIC ASSOCIATION/RECOGNITION** **Editorial Board** 1. Editorial Board Member, Universal Journal of Chemistry and Applications (previously - Journal of Chemistry and Applications), RAFT Publishers. 2. Editorial Board Member & Reviewer Board Member, International Journal of Advanced Research in Engineering & Technology (IJARET), International Association for Engineering and Management Education.

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		3. Editorial Board Member, International Journal of Environmental Science and Development (IJESD), (International Association of Computer Science and Information Technology Press, IACSIT) 4. International Advisory Editorial Board Member, Journal of Applied Research in Water and Wastewater (JARWW), Razi University, Iran 5. Editorial Board Member, Journal of Civil Engineering and Technology (JCIET), International Association for Engineering and Management Education. 6. (Previous/former) – Associate Editor, Journal of Environmental Science and Technology, Science Alert (Asian Network for Scientific Information). (Previous/former) – Associate Editor, Research Journal of Environmental Sciences, Science Alert (Academic Journals Inc.). 35 7. (Previous/former) – Editorial Board Member, International Journal of Civil Engineering and Technology (IJCIET), International Association for Engineering and Management Education. **PROFESSIONAL MEMBERSHIP** 1. Professional Technologist (Ts/P.Tech.), Malaysia Board of Technologists (MBOT) (Membership No. PT21110102) – (19 November 2021 to date). 2. Graduate Technologist, Malaysia Board of Technologists (MBOT) (Membership No. GT20090164) – (9 September 2020 to date). 3. Engineering Technologist, Board of Engineers Malaysia (BEM) (Registration No. T25824L) – (18 October 2018 to date). 4. Incorporated Member, The Institution of Engineers, Malaysia (IEM) (Membership Number: 94633) – (16 October 2017 to date). 5. Honorary Professorial Research Fellow, SEGi University College, Selangor (September 1, 2011 – August 31, 2014). 6. Senior Member, Hong Kong Chemical, Biological & Environmental Engineering Society (HKCBEES). (renamed, previously "Asia-Pacific Chemical, Biological & Environmental Engineering Society", APCBEES). (Membership Number: 100341) – (1 November, 2011 to date). 7. Member, Malaysia TRIZ Innovation Association – (8 June, 2011 to date). Member, Scientific and Technical Committee & Editorial Review Board on Engineering and Applied Sciences, World Academy of Science, Engineering and Technology (WASET) – (11 May, 2011 to date) 8. Member, International Association of Engineers (Member No: 107184) – (17 May, 2010 to date). **9.** (Previous/former) – Senior Associate Member, Institution of Chemical Engineers, IChemE - UK (Membership Number: 99945072) – (27 July 2010 – 31 August 2021). **ACADEMIC/COMMITTEE MEMBER** 1. Member, Sustainability committee, School of Chemical Engineering, Universiti Sains Malaysia (January 2, 2025 – December 31, 2027). 2. Member of Academic, Teaching and Learning Committee (Ahli Jawatankuasa Akademik, Pengajaran dan Pembelajaran), School of Chemical Engineering, Universiti Sains Malaysia (January 2, 2025 – December 31, 2027). 3. Sub-committee member (Publication & Promotion) (Ahli Jawatankuasa Kecil Publisiti dan Promosi) for 8th International Conference on Environment 2024 (ICENV2024), July 10, 2023.

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		4. Member of the working committee, Times Higher Education – Global Impact Ranking (THE GIR) – "Measuring Universities Contribution to The United Nation's Sustainable Developments Goals (SDGs)", SDG 14: Life Below Water, Universiti Sains Malaysia (August 5, 2022 – December 31, 2024) 5. Member, Sustainability committee, School of Chemical Engineering, Universiti Sains Malaysia (January 1, 2022 – December 31, 2024). 6. Member of Academic, Teaching and Learning Committee (Ahli Jawatankuasa Akademik, Pengajaran dan Pembelajaran), School of Chemical Engineering, Universiti Sains Malaysia (January 1, 2022 – December 31, 2024). 7. Technical Program Committee, in 2nd Asia Conference on Renewable Energy And Environmental Engineering (AREEE 2021), Online, March 27-29, 2021, Singapore. 8. Technical Committee, in 6th GoGreen Summit & Awareness on Climate Change, July 1- 2, 2020, Hotel Santika Premiere Malang, Malang, Indonesia. 9. Deputy chairman (Publication committee) for the public lecture of newly appointed professors (Timbalan Ketua Jawatankuasa Penerbitan Majlis Syarahan Umum 36 Pelantikan Profesor), School of Chemical Engineering, Universiti Sains Malaysia (November 6, 2019). **10.** Technical Program Committees, in 4th Asia Conference on Environment and Sustainable Development (ACESD 2019), November 9-11, 2019, The JICA Yokohama Center, Japan
15.	Maklumat Sasaran KPI Untuk Tujuan Pelantikan	**1. M1: Penerbitan / Publications** a. Minimum empat (4) penerbitan setahun - penerbitan dalam jurnal berindeks sebagai *First Author/Corresponding Author* *A minimum of four (4) publications per year - publications in indexed journals as the First Author/ Corresponding Author* **2. M1: Penyelidikan / Research** a. 1 geran jangka pendek / kebangsaan/industri/antarabangsa setahun *1 short term grant / national/industrial/international grant per year* **3. M2: Pengajaran / *Teaching*** a. Minimum 8 unit per sidang akademik *Minimum 8 units per academic session* **4. M2: Penyeliaan / Supervision** d. Penyelian 3 - 5 pelajar FYP dan 3 - 5 Latihan Industri per sidang *Supervision: 3-5 FYP students and 3-5 Industrial Training students per session* e. 1 pelajar pasca siswazah (Sarjana) / *1 postgraduate student (Master's)* f. 1 pelajar pasca siswazah (PhD) *1 postgraduate student (PhD)* **5. Penjanaan Pendapatan / Income Generation** a. Minimum 2 aktiviti penjanaan kewangan seperti konsultasi, kursus-kursus profesional dan lain-lain b. *Minimum of 2 financial generation activities such as consultations, professional courses, etc.*

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		Prestasi staf akan dinilai setiap tahun bagi memastikan sasaran KPI yang telah ditetapkan tercapai. *The performance of staff will be reviewed yearly to ensure the fulfillment of the designated KPI targets.*
16.	Penetapan Gaji	Gaji Semasa Gred DS13 : RM6,210.00 Imbuhan Tetap Keraian : RM600.00 Imbuhan Tetap Perumahan : RM700.00 Bayaran Bantuan Sara Hidup : RM300.00 **Jumlah Keseluruhan : RM7,810.00**
17.	Justifikasi Lantikan	1. Pelantikan ini adalah untuk memenuhi kekurangan tenaga pengajar dan memenuhi kekosongan bagi menggantikan satu (1) orang Pensyarah Universiti (Kontrak – Bukan Warganegara/Ekspatriat) di Bahagian Teknologi yang tamat perkhidmatan: ▪ Dr. Esam Bashir Abdulsalam Yahya (01.09.2022 - 23.04.2025) 2. Keperluan kepakaran dalam pembelajaran dan pengajaran. Bertanggungjawab dalam semua aspek yang berhubungan dengan pelaksanaan tugas pengajaran dan pembelajaran di Pusat Pengajian Teknologi Industri seperti yang diperlukan dan ditetapkan oleh PTJ. 3. Keperluan kepakaran dalam penyelidikan dan penyeliaan pelajar ijazah tinggi (sumbangan kepada penyelidikan, penyeliaan dan penerbitan) yang dapat membantu meningkatkan prestasi Bahagian, Pusat Pengajian dan Universiti. 4. Pelantikan ini dapat menyumbang kepakaran dalam meningkatkan prestasi dan KPI pusat pengajian. 5. Pelantikan ini akan dapat memastikan perancangan dan hala tuju pusat pengajian tidak terjejas disebabkan kekurangan tenaga pengajar dalam bidang ini. 6. Memenuhi bidang tujahan yang diperlukan.
18.	Perakuan Jawatankuasa Kecil PTJ	Setelah menemuduga calon, Jawatankuasa Kecil – Sumber Manusia berpuas hati dan bersetuju memperakukan pelantikan Dr. Ts. Irvan Dahlan Mansyoer sebagai Pensyarah Universiti DS13 (Kontrak – Bukan Warganegara) kerana memenuhi semua syarat dan keutamaan yang diperlukan serta justifikasi berikut: 1. Pelantikan ini adalah untuk memenuhi kekosongan jawatan di Bahagian Teknologi Bioproses, Pusat Pengajian Teknologi Industri. 2. Memenuhi bidang tujahan yang diperlukan. 3. Rekod penerbitan yang sangat baik. 4. Berpotensi untuk melonjak KPI PTJ. 5. Calon mempunyai pengalaman dalam pengajaran dan pembelajaran. Pelantikan ini bagi tempoh dua (2) tahun berkuat kuasa pada tarikh melapor diri.

2-B	NAMA	DR. TS. IRVAN DAHLAN MANSYOER (PILIHAN KE DUA)
		Nota: Perakuan ini adalah sebagai calon pilihan kedua (simpanan) dan tertakluk kepada semakan status calon dengan pihak Bahagian Sumber Manusia bagi urusan pelantikan staf kontrak yang telah lama bermastautin serta keputusan terdahulu universiti bagi pelantikan sedia ada beliau. Sekiranya hasil semakan tidak bersesuaian, calon pilihan ketiga akan dimajukan untuk pertimbangan Jawatankuasa Pemilih.
19.	Catatan Tambahan PTJ	**Tugasan Pengajaran dan Pembelajaran (P&P)** Antara kursus yang terlibat: 7. IEK 212/3 Process Heat Transfer 8. IEK 213/3 Mass Transfer and Separation Process 9. IEA 216/3 Computer Application in Industry 10. IBK 314 Downstream Process Technology 11. IBA407 Practical in Downstream Processing 12. IBG307 Bioprocess Optimization And Simulation
20.	Perakuan Jawatankuasa Pemilih	
21.	Nota Rujukan JKPTJSM	

LAMPIRAN C

3-C	NAMA	DR. MUHAMMAD ZUBAIR SHAHID (SIMPANAN - PILIHAN KEDUA)
1.	Umur	38 tahun (7 Oktober 1986)
2.	Warganegara	Pakistan
3.	Jabatan	Pusat Pengajian Teknologi Industri
4.	Kelayakan Akademik	PhD in Chemical Engineering Universiti PETRONAS Malaysia 2021 M.Sc. in Chemical Engineering Universiti PETRONAS Malaysia 2015 BSc in Chemical Engineering University of the Punjab, Lahore, Pakistan 2010
5.	H-Index 1. Scopus 2. WoS	h-index: 10
6.	Bidang Kepakaran	CO2 capture, process modelling, process analytics, mathematical modelling, experimental methods, MATLAB, Aspen Plus, Aspen HYSYS
7.	Pengalaman (Akademik / Industri /dan sebagainya)	**1.** Postdoctoral Fellow Interdisciplinary Research Center for Hydrogen Technologies and Carbon Management, King Fahad University, Kingdom of Saudi Arabia January 2024-Present 2. Postdoctoral Researcher Hanyang University, Seoul South Korea Chemical Engineering Department March 2022-December 2023 3. Part-time Graduate Assistant/Teaching Assistant Universiti Teknologi PETRONAS, Malaysia Chemical Engineering Department March 2013-October 2015 & July 2016-April 2021 **INDUSTRIAL EXPERIENCE/TRAININGS** ▪ Worked as a Technical Support Engineer ▪ Mal Pakistan Limited, Pakistan (July 2011-September 2011) ▪ Internee in the research and development department ▪ MB Dyes and Chemicals Private Limited, Pakistan (July 2010-August 2011) ▪ Internee at the process plant • Rupafil Polyester Limited, Pakistan (July 2008-August 2008)

3-C	NAMA	DR. MUHAMMAD ZUBAIR SHAHID (SIMPANAN - PILIHAN KEDUA)
8.	Senarai Pencapaian Penyelidikan dan Penerbitan (untuk tempoh 5 tahun terkini)	**PUBLICATION/JOURNAL** **2025** 1. Sustainable Aviation Fuel: Operational Challenges, Techno-economics, and Life Cycle Analysis Mhadi A Ismael, Mohammed El-Adawy, Ahmad Salam Farooqi, Mohamed Hamdy, Muhammad Zubair Shahid, Zeineldin Elserfy, Medhat A Nemitallah Energy & Fuels, 2025 2. Effects of CaO/MgO Compositions on the Performances of Sorption Enhanced Water Gas Shift Reaction: A Thermodynamic Analysis Muhammad Zubair Shahid, Ahmad Salam Farooqi, Mohammad Mozahar Hossain, Usama Ahmed Chemical Engineering and Processing-Process Intensification, 2025, 110287 3. Performance of Ni-M/Ca12Al14O33 (M= Co, Cu and Fe) bimetallic catalysts in sorption-enhanced steam methane reforming for blue hydrogen production Ahmad Salam Farooqi, Muhammad Zubair Shahid, Mohamed Essalhi, Mohammad M Hossain, Mahmoud M Abdelnaby, Mohammed A Sanhoob, Vasilije Manovic, Medhat A Nemitallah Catalysis Today, 2025, 115281 4. Clean hydrogen production via sorption enhanced water gas shift reaction: A comprehensive review Muhammad Zubair Shahid, Ahmad Salam Farooqi, Kevin Fajri, Mohammed El-Adawy, Mohamed Hamdy, Abid Salam Farooqi, Omar Y. Abdelaziz, Mohammad Mozahar Hossain, Medhat A. Nemitallah International Journal of Hydrogen Energy, 2025, 483-1512 5. Advancements in sorption-enhanced steam reforming for clean hydrogen production: A comprehensive review Ahmad Salam Farooqi a Abdelwahab N. Allam , Muhammad Zubair Shahid , Anas Aqil, Kevin Fajri , Sunhwa Park, Omar Y. Abdelaziz, Mahmoud M. Abdelnaby, Mohammad M. Hossain, Mohamed A. Habib, Syed Muhammad Wajahat ul Hasnain, Ali Nabavi, Mingming Zhu, Vasilije Manovic , Medhat A. Nemitallah Carbon Capture Science & Technology, 2025, 100336 6. Recent advances of in-well hydrogen production with integrated in-situ carbon dioxide sequestration: A comprehensive review Mohamed Hamdy, Muhammad Zubair Shahid, Ahmad Salam Farooqi, Mohammed El-Adawy, Mohamed Mahmoud, Sulaiman Alarifi, Medhat A Nemitallah Process Safety and Environmental Protection, 2025, 1387-1408 **2024** 1. Removal of Carbon Dioxide and Hydrogen Sulfide from Natural Gas Using a Hybrid Solvent of Monoethanolamine and N-Methyl 2-Pyrrolidone Abid Salam, Farooqi Raihan Mahirah Ramli,Serene Sow Mun Lock, Ahmad Salam Farooqi, Muhammad Zubair Shahid, Syed Muhammad Wajahat ul Hasnain, Noor e Hira, Bawadi Abdullah ACS Omega 2024, 9, 24, 25704–25714.

3-C	NAMA	DR. MUHAMMAD ZUBAIR SHAHID (SIMPANAN - PILIHAN KEDUA)
		2. Design and economic evaluation of a novel amine-based CO2 capture process for SMR-based hydrogen production plants. Muhammad Zubair Shahid, Jin-Kuk Kim Journal of Cleaner Production 402, 136704. **2022** 1. CO2 Absorption from Biogas Using Piperazine-Promoted 2-Amino-2-methyl-1-propanol: Process Performance in a Packed Column. Viga Rajiman, Hairul Nazirah Abdul Halim, Azmi Mohd Shariff, Muhammad Zubair Shahid, Abdulhalim Shah Maulud, Kok Keong Lau and Lian See Tan Sustainability 2022, 14(12), 7095. 2. Simulation of Natural Gas Treatment for Acid Gas Removal Using the Ternary Blend of MDEA, AEEA, and NMP. Abid Salam Farooqi, Raihan Mahirah Ramli, Serene Sow Mun Lock, Noorhidayah Hussein, Muhammad Zubair Shahid and Ahmad Salam Farooqi Sustainability 2022, 14(17), 10815. **2021** 1. Raman Calibration Models for Chemical Species Determination in CO2-Loaded Aqueous MEA Solutions Using PLS and ANN Techniques. Ahmad Syukri Hanafiah Abdulhalim Shah Maulud, Muhammad Zubair Shahid, Humbul Suleman, and Azizul Buang ChemEngineering 2021, 5(4), 87. 2. Development of an extended model for the permeation of environmentally hazardous CO2 gas across asymmetric hollow fiber composite membranes. Asif Jamil, Oh Pei Ching, Tanveer Iqbal, Sikander Rafiq, Muhammad Zia-ul-Haq, Muhammad Zubair Shahid, Muhammad Mubashir, Sivakumar Manickam, Pau Loke Show Journal of Hazardous Materials, 126000, 2021. 3. Packed column modelling and experimental evaluation for CO2 absorption using MDEA solution at high pressure and high CO2 concentration. MZ Shahid, AS Maulud, MA Bustam, H Suleman, HNA Halim, AM Shariff Journal of Natural Gas Science and Engineering 88, 103829, 2021. 4. A review of semi-empirical equilibrium models for CO2-alkanolamine-H2O solutions and their mixtures at high pressure. H Suleman, AS Maulud, PL Fosbøl, Q Nasir, R Nasir, MZ Shahid Journal of Environmental Chemical Engineering, 104713, 2021. **2020** 1. The contribution of aqueous l-arginine salts to equilibrium carbon dioxide absorption in a co-promoter role at high pressure. H Suleman, AS Maulud, A Syalsabila, MZ Shahid Fluid Phase Equilibria 524, 112743, 2020. 2. Densities and Thermal Expansion Coefficients of Pure Methylcyclohexane, 1-

3-C	NAMA	DR. MUHAMMAD ZUBAIR SHAHID (SIMPANAN - PILIHAN KEDUA)
		Methylcyclohexene, 4-Methylcyclohexene, 1-Methyl-1, 4-cyclohexadiene, and Toluene and Binary Mixtures. MR Usman, Z Shahid, MS Akram, R Aslam International Journal of Thermophysics 41 (4), 1-17, 2020. 3. High-pressure experimental and theoretical study of CO2 solubility in aqueous blends of lysine salts with piperazine as new absorbents. H Suleman, AS Maulud, A Syalsabila, MZ Shahid, PL Fosbøl Fluid Phase Equilibria 507, 112429, 2020. **INTERNATIONAL CONFERENCE/PROCEEDINGS** 1. Removal of CO2 from PSA tail gas stream in the SMR hydrogen production lant using MEA solvent at high pressure MZ Shahid, MA Nemitallah, AS Farooqi, MM Hossain CSChE 2024, 74th Canadian Chemical Engineering Conference, Toronto, Canada 2 Impact of thermophysical and transport properties of amine-based solvents on the economy of the CO 2. Absorption process 주비르(MZ Shahid), 김진국 Korean Society of Industrial Chemistry, 2023 3. Application of CO2-MDEA absorption process to capture CO2 from PSA tail gas in SMR hydrogen production plant 주비르 (MZ Shahid), 이주화 , 박하린 , 김진국 Korean Society of Industrial Chemistry, 2022 4. Modeling of CO2-MEA absorption system in the packed column using Sulzer DX structured packing. MZ Shahid, AS Maulud, MA Bustam, H Suleman IOP Conference Series: Materials Science and Engineering 736 (2), 022059, 2020
9.	Senarai Penglibatan Pengajaran dan Penyeliaan (untuk tempoh 5 tahun terkini)	-
10.	Senarai Pencapaian Sanjungan Akademik dan Anugerah (untuk tempoh 5 tahun terkini)	1. Secured graduate assistantship during PhD studies. 2. Secured graduate assistantship during MSc studies. Won a bronze medal in a research exhibition (SEDEX 32) at University Teknologi PETRONAS, Malaysia.
11.	Senarai Penglibatan Konsultansi (untuk tempoh 5 tahun terkini)	-
12.	Senarai Penglibatan Sumbangan Kepada Universiti dan Komuniti (untuk tempoh 5 tahun terkini)	-
13.	Maklumat Tambahan Calon (Sekiranya Ada)	**SKILLS SUMMARY** • Teaching • Blue hydrogen production technologies • Process analytics • Designing experimental systems

3-C	NAMA	DR. MUHAMMAD ZUBAIR SHAHID (SIMPANAN - PILIHAN KEDUA)
		• Pilot-scale experimentation • Mathematical and process modelling • MATLAB programming • Aspen HYSYS and Aspen Plus • Raman spectroscopy • Adsorption based CO2 capture • Synthesis for metallic catalysts **COMPUTER SOFTWARE SKILLS** • Level Skills • Advanced MATLAB, Aspen HYSYS, Aspen Plus, HSC-Chemistry • Basic COMSOL Multiphysics, COMSO-RS **CO-CURRICULAR ACTIVITIES** • 3rd position in a presentation competition held at the University of Engineering and Technology, Lahore. • Best speaker award in Debates and Speeches Competition at College • Best speaker award in Debates and Speeches Competition in School. • School badminton and cricket team player **CERTIFICATIONS/TRAINING** • AI Training Workshop for Chemical Engineers at Canadian Chemical Engineering Conference (CSChe 2024) in Totonto, ON, Canada. • Introduction to Process Modeling using Aspen HYSYS (Aspentech) (July 2022) • Laboratory Safety Training Course (December 2018) • A short course on introduction to COSMO-RS-Model-Theory and practices (March 2015) • A short course on mastering the art of publication-A versatile approach (February 2015) • Raman Spectroscopy advance training (August 2014) • Outcome-based education (OBE) for technologists and graduate assistants (July 2014) • Professional development seminar on “Aspen Plus for Process Design and Simulation” (February 2012) • Short course on AutoCAD (April 2008)
14.	Maklumat Sasaran KPI Untuk Tujuan Pelantikan	**1. M1: Penerbitan / Publications** a. Minimum empat (4) penerbitan setahun - penerbitan dalam jurnal berindeks sebagai *First Author/Corresponding Author* *A minimum of four (4) publications per year - publications in indexed journals as the First Author/ Corresponding Author* **2. M1: Penyelidikan / Research** a. 1 geran jangka pendek / kebangsaan/industri/antarabangsa setahun *1 short term grant / national/industrial/international grant per year* **3. M2: Pengajaran / *Teaching*** a. Minimum 8 unit per sidang akademik *Minimum 8 units per academic session* **4. M2: Penyeliaan / Supervision** g. Penyelian 3 - 5 pelajar FYP dan 3 - 5 Latihan Industri per sidang

3-C	NAMA	DR. MUHAMMAD ZUBAIR SHAHID (SIMPANAN - PILIHAN KEDUA)
		Supervision: 3-5 FYP students and 3-5 Industrial Training students per session h. 1 pelajar pasca siswazah (Sarjana) / *1 postgraduate student (Master's)* i. 1 pelajar pasca siswazah (PhD) *1 postgraduate student (PhD)* **5. Penjanaan Pendapatan / Income Generation** a. Minimum 2 aktiviti penjanaan kewangan seperti konsultasi, kursus-kursus profesional dan lain-lain b. *Minimum of 2 financial generation activities such as consultations, professional courses, etc.* Prestasi staf akan dinilai setiap tahun bagi memastikan sasaran KPI yang telah ditetapkan tercapai. *The performance of staff will be reviewed yearly to ensure the fulfillment of the designated KPI targets.*
15.	Penetapan Gaji	Gaji Semasa Gred DS51 : RM6,210.00 Imbuhan Tetap Keraian : RM600.00 Imbuhan Tetap Perumahan : RM700.00 Bayaran Bantuan Sara Hidup : RM300.00 **Jumlah Keseluruhan : RM7,810.00**
16.	Justifikasi Lantikan	1. Pelantikan ini adalah untuk memenuhi kekurangan tenaga pengajar dan memenuhi kekosongan bagi menggantikan satu (1) orang Pensyarah Universiti (Kontrak – Bukan Warganegara/Ekspatriat) di Bahagian Teknologi yang tamat perkhidmatan: ▪ Dr. Esam Bashir Abdulsalam Yahya (01.09.2022 - 23.04.2025) 2. Keperluan kepakaran dalam pembelajaran dan pengajaran. Bertanggungjawab dalam semua aspek yang berhubungan dengan pelaksanaan tugas pengajaran dan pembelajaran di Pusat Pengajian Teknologi Industri seperti yang diperlukan dan ditetapkan oleh PTJ. 3. Keperluan kepakaran dalam penyelidikan dan penyeliaan pelajar ijazah tinggi (sumbangan kepada penyelidikan, penyeliaan dan penerbitan) yang dapat membantu meningkatkan prestasi Bahagian, Pusat Pengajian dan Universiti. 4. Pelantikan ini dapat menyumbang kepakaran dalam meningkatkan prestasi dan KPI pusat pengajian. 5. Pelantikan ini akan dapat memastikan perancangan dan hala tuju pusat pengajian tidak terjejas disebabkan kekurangan tenaga pengajar dalam bidang ini. 6. Memenuhi bidang tujahan yang diperlukan.

3-C	NAMA	DR. MUHAMMAD ZUBAIR SHAHID (SIMPANAN - PILIHAN KEDUA)
17.	Perakuan Jawatankuasa Kecil PTJ	Setelah menemuduga calon, Jawatankuasa Kecil – Sumber Manusia berpuas hati dan bersetuju memperakukan pelantikan Dr. Muhammad Zubair Shahid sebagai Pensyarah Universiti DS13 (Kontrak – Bukan Warganegara) kerana memenuhi semua syarat dan keutamaan yang diperlukan serta justifikasi berikut: 1. Pelantikan ini adalah untuk memenuhi kekosongan jawatan di Bahagian Teknologi Bioproses, Pusat Pengajian Teknologi Industri. 2. Memenuhi bidang tujahan yang diperlukan. 3. Rekod penerbitan yang sangat baik. 4. Berpotensi untuk melonjak KPI PTJ. 5. Calon mempunyai pengalaman dalam pengajaran dan pembelajaran. Pelantikan ini bagi tempoh dua (2) tahun berkuat kuasa pada tarikh melapor diri. Nota: Perakuan ini adalah sebagai calon simpanan pilihan kedua/ketiga.
18.	Catatan Tambahan PTJ	**<u>Tugasan Pengajaran dan Pembelajaran (P&P)</u>** Antara kursus yang terlibat: 1. IEK 212/3 Process Heat Transfer 2. IEK 213/3 Mass Transfer and Separation Process 3. IEA 216/3 Computer Application in Industry 4. IBK 314 Downstream Process Technology 5. IBA407 Practical in Downstream Processing 6. IBG307 Bioprocess Optimization And Simulation
19.	Perakuan Jawatankuasa Pemilih	
20.	Nota Rujukan JKPTJSM	

LAMPIRAN D

4-D	NAMA	DR. PADMESH TIRUNELVELI NARAYANAPILLAI
1.	Umur	46 tahun (14 November 1979)
2.	Warganegara	India
3.	Jabatan	Pusat Pengajian Teknologi Industri
4.	Kelayakan Akademik	PhD in Chemical Engineering – Wastewater Treament Anna University 2006 Master of Engineering (Chemical Engineering) Annamalai University 2002 Bachelor of Engineering (Chemical Engineering) University of the Punjab, Lahore, Pakistan 2001
5.	H-Index 1. Scopus 2. WoS	H – Index: 10 Citations: 1,394 Scopus ID: 8707103800
6.	Bidang Kepakaran	▪ Environmental Biotechnology: Specializing in the biological treatment of pollutants like dyes and heavy metals, waste management, and sustainable environmental solutions. ▪ Chemical Engineering Core: Strong knowledge in fundamental chemical engineering principles including fluid mechanics, heat and mass transfer, reaction engineering, and process control. ▪ Process Design & Simulation: Proficient in chemical process design, safety, and simulation tools. ▪ Sustainable Energy: Experience in developing biofuels (biogas, biodiesel) from renewable sources. ▪ Nanomaterials: Expertise in the green synthesis of nanoparticles. ▪ Research & Academic Leadership: Proven ability in research supervision, project guidance, curriculum development, and extensive teaching experience in chemical and environmental engineering. ▪ Analytical & Computing: Skilled in various analytical techniques and software relevant to engineering and research.
7.	Pengalaman (Akademik / Industri /dan sebagainya)	Assistant Professor University : Mila University, Malaysia. (Formerly known as Manipal International University) (October 2023 – Present Assistant Professor University : Manipal International University, Malaysia. (June 2014 – October 2023) Senior Lecturer (July 2011 – January 2014) & Lecturer (May 2009 – July 2011) Taylor's University, Subang Jaya, Malaysia.

4-D	NAMA	DR. PADMESH TIRUNELVELI NARAYANAPILLAI
		Lecturer (June 2006 – May 2008) & Senior Lecturer (June 2008 – May 2009) University : Sathyabama University, Chennai – 119, India
8.	Senarai Pencapaian Penyelidikan dan Penerbitan (untuk tempoh 5 tahun terkini)	**PUBLICATION/JOURNAL** **2024** 1. Nair Amish, Tirunelveli Narayanapillai Padmesh and Selvaraju Kishanraj (2024) "Bio-Enzymes in the Detergent Industry: A Pathway for Sustainable Entrepreneurship through Waste Utilization" Discover Sustainability Journal, Special Issue Springer (Under Review). 2. T.V.N. Padmesh (2024) Batch and Column biosorption of acid dyes from synthetic solutions onto Crab shell, Envin. Polln. Journal (Under Review). 3. T.V.N. Padmesh, (2024) Biosorption potential of Azolla on industrial bearing dye effluents: Batch and Column studies, Algae Research (Under Review). **4.** Darshaleka Chellappa, Soundarajan Krishnan and Padmesh Tirunelveli Narayana Pillai (2024) "Biosorption of Methylene Blue (MB) dye using agricultural waste Parkia speciosa: One factor at a time method (OFAT), Isotherm and Kinetic Studies" Vol 10 (4) pp. 235-249 https://doi.org/10.1504/IJNBM.2024.143830 (Scopus Indexed). **2023** 1. Padmesh Tirunelveli Narayana Pillai, Boon Chee Yong (2023) "Aspen adsorption simulation on biosorption between water hyacinth (Eichhornia crassipes) and Pb (II) ions in packed bed column" IOP Conference Series: Materials Science and Engineering. 1257 012049 {DOI 10.1088/1757-899X/1257/1/012049). **2.** G. Narayanasamy, TNP. Padmesh, Soundarajan Krishnan (2023) "Experimental Studies on Transesterification of Mango (Mangifera indica) Seed Kernel Butter for Biodiesel Production" IOP Conference Series: Materials Science and Engineering. 1257 012012 (DOI 10.1088/1757-899X/1257/1/012012). **BOOK CHAPTERS** 1. Tirunelveli Narayanapillai Padmesh*, Indu Thanammal, Velu Vengadeshwaran (2025) "Challenges and Ethical Considerations," for the forthcoming edited volume, Role of Artificial Intelligence in Higher Education: Pathways to Sustainable Development, to be published by Springer Nature (Scopus Indexed) – Submitted – Under Review. **2.** Tirunelveli Narayanapillai Padmesh*, Velu Vengadeshwaran, Indu Thanammal, Suguna Selvakumaran, Ong Ying Hui (2025) "Bioenergy Systems: Challenges and Opportunities for a Sustainable Future," for the upcoming book, "Harnessing Biodiversity for a Sustainable Bioeconomy", IGI Global Scientific Publishing – Scopus Indexed – Under Review. **INTERNATIONAL CONFERENCE PRESENTATIONS** 1. Nair Amish, Tirunelveli Narayanapillai Padmesh and Selvaraju Kishanraj (2024) "Bio-Enzymes in the Detergent Industry: A Pathway for Sustainable Entrepreneurship through Waste Utilization" SME

4-D	NAMA	DR. PADMESH TIRUNELVELI NARAYANAPILLAI
		World Forum Kuala Lumpur 2024 (SWFKL24) 29 November 2024, | MILA University, Malaysia. 2. Darshaleka Chellappa, Soundarajan Krishnan and Padmesh Tirunelveli Narayanapillai (2023) "Biosorption of Methylene Blue (MB) dye using agricultural waste Parkia speciosa: One factor at a time method (OFAT), Isotherm and Kinetic Studies" 2nd International Conference on Innovation in Chemical Engineering and Technology (ICICET2023) 4 - 5 September 2023 | Universiti Putra Malaysia, Malaysia. 3. Vaigeswary N. Rammu and Padmesh Tirunelveli Narayanapillai (2023) "Review studies on the design and analysis of honeycomb-based nonpneumatic tyres" 2nd International Conference on Innovation in Chemical Engineering and Technology (ICICET2023) 4 - 5 September 2023 | Universiti Putra Malaysia, Malaysia. 4. Rashvain Nana Seakaran and Padmesh Tirunelveli Narayanapillai (2023) "Review and comparative analysis of technological upgrades in the 48V mild hybrid technology and their economic and environmental impact" 2nd International Conference on Innovation in Chemical Engineering and Technology (ICICET2023) 4 - 5 September 2023 | Universiti Putra Malaysia, Malaysia. 5. Padmesh Tirunelveli Narayanapillai, Nisha Fareena (2022) "Artificial Intelligence to Predict the Production of Hydrogen Through Photoreforming of Polyethylene Terephthalate Waste Using Cnx|Ni2p Photocatalyst" 19th Asian Pacific Confederation of Chemical Engineering Congress (APCChE 2022). 9 - 12 Aug 2022, Kuala Lumpur, Malaysia. **6.** Padmesh Tirunelveli Narayanapillai, Boon Chee Yong (2022) "A Study on Dynamic Simulation of Pb (II) Biosorption in Eichhornia Crassipes Packed Bed Column" 33rd Symposium of Malaysian chemical engineers 2022 (SOMCHE 2022). Malaysia 7. G. Narayanasamy, TNP. Padmesh, Soundarajan Krishnan (2022) "Experimental Studies on Transesterification of Mango (Mangifera indica) Seed Kernel Butter for Biodiesel Production" 33rd Symposium of Malaysian chemical engineers 2022 (SOMCHE 2022). Malaysia 8. Predaarshan V. Chandran, Padmesh T.V.N and Soundarajan Krishnan (2020) "Studies on Water Defluoridation using Acid-Activated Kaolinite Clay", The 2nd International Conference on Contemporary Issues' 2020, ICCI' 20, Malaysia. **9.** Thivagaran Anand, Padmesh T.V.N and Soundarajan Krishnan (2020) "Biosorption of Chromium (III) Ions from Aqueous Solution using NaOH-treated and HCl-treated Activated Carbon from Kiwi Fruit Peels", The 2nd International Conference on Contemporary Issues' 2020, ICCI' 20, Malaysia. 10. Raannesh Jeyakumar and Padmesh T.V.N (2020) "ASPEN Simulation on Biosorption of Brilliant Orange Dye on to Activated Carbon Axonopus compressus", The 2nd International Conference on Contemporary Issues' 2020, ICCI' 20, Malaysia. 11. Chudersen Nana Seakaran, Padmesh T.V.N and Soundarajan Krishnan (2020) "Studies on Removal of Copper (II) Ions from Aqueous Solution using Acid-Treated and Alkali-Treated Powdered

4-D	NAMA	DR. PADMESH TIRUNELVELI NARAYANAPILLAI
		Banana Peels", The 2nd International Conference on Contemporary Issues' 2020, ICCI' 20, Malaysia. 12. Nagheshwari Gnanasekharan and TVN. Padmesh (2020) "Batch Biosorption of Mixed Dyes on to Duckegg Shell: Equilibrium Isotherm Studies", The 2nd International Conference on Contemporary Issues' 2020, ICCI'20, Malaysia. 13. Jemimah Caren Thiyagarajan and TVN. Padmesh (2020) "Studies on Removal of Calcium Carbonate from Wastewater Using Citric Acid from Lemon Peel Powder", The 2nd International Conference on Contemporary Issues' 2020, ICCI' 20, Malaysia. 14. Sangeetha Kartigesu and TVN. Padmesh (2020) "Biosorption of Brilliant Red by Eggshell in Batch Reactor Using Response Surface Methodology", The 2nd International Conference on Contemporary Issues' 2020, ICCI' 20, Malaysia. 15. Dhaarmendraa Tamilselvan and TVN. Padmesh (2020) "Column Studies on Removal of Mixed Dyes in Aqueous Solution by Using Duck Eggshells as Biosorbent", The 2nd International Conference on Contemporary Issues' 2020, ICCI' 20 Malaysia. **16.** Hemapriyankaa Periyathamby and TVN. Padmesh (2020) "Comparison of Duck and Chicken Feather on Biosorption of Brilliant Blue Dye on Fixed Bed Column Using Response Surface Methodology and Anova", The 2nd International Conference on Contemporary Issues' 2020, ICCI' 20 Malaysia. **RESEARCH ACCOMPLISHMENTS** 1. Identified and isolated bacterial species present in industrial waste, successfully applying them for the degradation of mono- and polyaromatic hydrocarbons, as well as azo dyes. 2. Conducted a comprehensive analysis of dye industry effluents, achieving complete removal of dyes from two industrial effluents using a biosorbent-packed column. 3. Engineered and constructed packed columns tailored for the treatment of authentic industrial effluents. 4. Developed a nonlinear controller using MATLAB software to regulate liquid levels in a nonlinear tank system. 5. Optimized culturing conditions for algae and devised a photobioreactor for the efficient production of biodiesel. **6.** Explored power generation possibilities from biomass sources such as vegetable waste and Palm EFB (Empty Fruit Bunches).
9.	Senarai Penglibatan Pengajaran dan Penyeliaan (untuk tempoh 5 tahun terkini)	**UNDERGRADUATE PROJECTS HANDLED** **<u>Final Year Projects</u>** **Manipal International University** **March 2019** • A Bio-coagulant Pisum sativum on Industrial Wastewater Electrochemical Method—Its Parametric Effects • Column Studies on Removal of Mixed Dyes in Aqueous Solution using Duck Eggshells as Biosorbent

4-D	NAMA	DR. PADMESH TIRUNELVELI NARAYANAPILLAI
		• Studies on Removal of Calcium Carbonate from Wastewater using Citric Acid from Lemon Peel Powder • Studies on Removal of Mixed Dyes from Aqueous Solution using Eggshell as Biosorbent • Studies on Removal of Mixed Heavy Metal Ions from Aqueous Solutions using Crab Shell as Biosorbent • Treatment of Model Turbid Water by Potential Evaluation of Cactus (Opuntia ficus indica) as Bio-coagulant • Chicken Eggshell as Effective Natural Adsorbent for the Removal of Dye from Synthetic Wastewater in Batch Reactor using Response Surface Methodology • Fixed Bed Bio-sorption of Dye from Textile Wastewater Effluent onto Duck and Chicken Feathers: Optimization Study using Response Surface Methodology **October 2019** • Magnetically Modified Biomass as a Biosorbent for Ionic Copper Removal • Magnetically Modified Biomass as a Biosorbent for Dye Removal • Optimization Studies on Removal of Nickel Ions (Ni++) from Aqueous Solution by Adsorption using Various Acid and Alkali-Treated Banana Peel Powder • Studies on Biosorption of Chromium Ions (Cr+++) from Aqueous Solution using NaOH and HCl-Treated Activated Carbon from Kiwi Fruit Peels • Optimization Studies on Removal of Copper Ions (Cu++) from Aqueous Solution using Acid- and Alkali-treated Powdered Banana Peels • Adsorption of Chromium using Silica Aerogel Nanoparticles **March 2020** • Biosorption Studies of Acid Dye onto to Acidic Modified Punica granatum L Peel • Biosorption Studies of Acid Dye onto to Nonpolar Modified Punica granatum L peel • Biosorption Studies of Acid Dye onto to Polar Modified Punica granatum L peel **October 2020** • Production of Biodiesel from Waste Green Apple Seed Oil • Production of Hydrogen Gas through Photo-reforming of Non-recyclable Plastic Waste using a Photocatalysts • Production of Biodiesel from Wood Waste **March 2021** • Biosorption: A Study on Dynamic Simulation of Pb(II) Adsorption in Packed Bed Column by Using Water Hyacinth (Eichhornia Crassipes) through ASPEN Adsorption **October 2021** • Removal of acid dyes by algae derived biosorbent. • Removal of acid dyes by agricultural derived biosorbent

4-D	NAMA	DR. PADMESH TIRUNELVELI NARAYANAPILLAI
		March 2022 • Optimizing biosorption of textile dyes using agricultural wastes using Artificial intelligence (AI) **October 2022** • Simulated Biosorption of heavy metals in single and binary metal systems by natural biosorbent using Aspen Adsorption **March 2023** • Simulated Biosorption of Copper & Nickel Ions using Litchi Chinensis sonn Peel in Single and Binary Metal Systems using Aspen Adsorption **October 2023** • Process simulation for biodiesel production from Jarak Belanda waste via aspen plus. • Simulation for biogas production process from organic waste via aspen plus.
10.	Senarai Pencapaian Sanjungan Akademik dan Anugerah (untuk tempoh 5 tahun terkini)	• "BEST TEACHER AWARD": Received during 2006-2007. • "YOUNG SCIENTIST PAPER AWARD": Received in 2010.
11.	Senarai Penglibatan Konsultansi (untuk tempoh 5 tahun terkini)	-
12.	Senarai Penglibatan Sumbangan Kepada Universiti dan Komuniti (untuk tempoh 5 tahun terkini)	▪ Curriculum Development: Developed the proposal and curriculum for a Master of Engineering Science program. ▪ Post Graduate Coordinator: Served as a Post Graduate Coordinator, contributing to the academic administration and management of postgraduate programs. ▪ Industrial Training Coordinator: Acted as an Industrial Training Coordinator, facilitating valuable industry exposure for students. ▪ Research Cluster Head: Led the "Environmental Research Group" as a Research Cluster Head, fostering research initiatives within the university. ▪ Journal Executive Editor: Served as Executive Editor for the Journal of Environment and Biotechnology Research (JEBR) and Journal of

4-D	NAMA	DR. PADMESH TIRUNELVELI NARAYANAPILLAI

		Engineering Science and Technology (JESTEC), contributing to academic publishing and dissemination of knowledge.
		• Conference Organization: Chaired two International level conferences (EURECA - International Conference), demonstrating leadership in organizing academic events and fostering collaboration.
		• Grant Management Committee Member: Contributed to the Research Management Committee as a member of the Grant Management Committee.
		• Invited Speaker/Guest Lecturer: Delivered talks at various institutions, sharing expertise on topics such as:
		  • "Research towards Sustainable Development Goals"
		  • "Waste Minimization and Management - Oil and Gas Industry"
		  • "Industry 4.0 in Chemical Engineering"
		  • "Outcome Based Education (OBE) in Engineering"
		  • "AI for Chemical Engineering Practices"
		Journal Reviewer: Reviewed articles for numerous international peer-reviewed journals, contributing to the quality and integrity of scientific publications in your field.

13.	Maklumat Tambahan Calon (Sekiranya Ada)	**PROFESSIONAL MEMBERSHIP** 1. Chartered Engineer (CEng), Member of Institution of Engineering and Technology (IET), UK. Membership No: 1100275788 **2.** Graduate Engineer Board of Engineers Malaysia Registration Number: G21128L **COMPUTER EXPERTISE** 1. Proficient in Microsoft Office suite 2. Experienced with MATLAB / Simulink 3. Familiar with Aspen Hysys Undergraduate Projects Handled: 4. Familiar in C Programming 5. Experienced with Minitab **6.** Beginner in Python programming **ANALYTICAL TECHNIQUES** 1. Measurement of Physical and Chemical Parameters: Including analysis of Biological Oxygen Demand (BOD), Chemical Oxygen Demand (COD), effluent properties, and determination of cations and anions using titration methods. 2. Instrumentation: Proficiency in operating Atomic Absorption Spectrophotometers (AAS) and UV-visible Spectrophotometers. 3. Spectroscopic Analysis: Utilizing Fourier Transform Infrared (FT-IR) spectroscopy and interpreting spectral peaks. Analytical Instrumentation: Experience with Flame Photometers, Water Analyzers, Gas Analyzers, Gas Chromatography-Mass Spectrometry (GC-MS), and High-Performance Liquid Chromatography (HPLC).
14.	Maklumat Sasaran KPI Untuk Tujuan Pelantikan	**1. M1: Penerbitan / Publications** a. Minimum empat (4) penerbitan setahun - penerbitan dalam jurnal berindeks sebagai *First Author/Corresponding Author* *A minimum of four (4) publications per year - publications in indexed journals as the First Author/ Corresponding Author*

4-D	NAMA	DR. PADMESH TIRUNELVELI NARAYANAPILLAI
		2. M1: Penyelidikan / Research a. 1 geran jangka pendek / kebangsaan/industri/antarabangsa setahun *1 short term grant / national/industrial/international grant per year* **3. M2: Pengajaran / *Teaching*** a. Minimum 8 unit per sidang akademik *Minimum 8 units per academic session* **4. M2: Penyeliaan / Supervision** j. Penyelian 3 - 5 pelajar FYP dan 3 - 5 Latihan Industri per sidang *Supervision: 3-5 FYP students and 3-5 Industrial Training students per session* k. 1 pelajar pasca siswazah (Sarjana) / *1 postgraduate student (Master's)* l. 1 pelajar pasca siswazah (PhD) *1 postgraduate student (PhD)* **5. Penjanaan Pendapatan / Income Generation** a. Minimum 2 aktiviti penjanaan kewangan seperti konsultasi, kursus-kursus profesional dan lain-lain b. *Minimum of 2 financial generation activities such as consultations, professional courses, etc.* Prestasi staf akan dinilai setiap tahun bagi memastikan sasaran KPI yang telah ditetapkan tercapai. *The performance of staff will be reviewed yearly to ensure the fulfillment of the designated KPI targets.*
15.	Penetapan Gaji	Gaji Semasa Gred DS51 : RM6,210.00 Imbuhan Tetap Keraian : RM600.00 Imbuhan Tetap Perumahan : RM700.00 Bayaran Bantuan Sara Hidup : RM300.00 **Jumlah Keseluruhan : RM7,810.00**
16.	Justifikasi Lantikan	1. Pelantikan ini adalah untuk memenuhi kekurangan tenaga pengajar dan memenuhi kekosongan bagi menggantikan satu (1) orang Pensyarah Universiti (Kontrak – Bukan Warganegara/Ekspatriat) di Bahagian Teknologi yang tamat perkhidmatan: ▪ Dr. Esam Bashir Abdulsalam Yahya (01.09.2022 - 23.04.2025) 2. Keperluan kepakaran dalam pembelajaran dan pengajaran. Bertanggungjawab dalam semua aspek yang berhubungan dengan pelaksanaan tugas pengajaran dan pembelajaran di Pusat Pengajian Teknologi Industri seperti yang diperlukan dan ditetapkan oleh PTJ. 3. Keperluan kepakaran dalam penyelidikan dan penyeliaan pelajar ijazah tinggi (sumbangan kepada penyelidikan, penyeliaan dan penerbitan) yang dapat membantu meningkatkan prestasi Bahagian, Pusat Pengajian dan Universiti.

4-D	NAMA	DR. PADMESH TIRUNELVELI NARAYANAPILLAI
		4. Pelantikan ini dapat menyumbang kepakaran dalam meningkatkan prestasi dan KPI pusat pengajian. 5. Pelantikan ini akan dapat memastikan perancangan dan hala tuju pusat pengajian tidak terjejas disebabkan kekurangan tenaga pengajar dalam bidang ini. 6. Memenuhi bidang tujahan yang diperlukan.
17.	Perakuan Jawatankuasa Kecil PTJ	Setelah menemuduga calon, Jawatankuasa Kecil – Sumber Manusia berpuas hati dan tidak memperakukan pelantikan Dr. Padmesh Turuneveli Narayanapillai sebagai Pensyarah Universiti DS13 (Kontrak – Bukan Warganegara) kerana pengalaman dan pencapaian beliau kurang bersesuaian dengan keperluan PTJ.
18.	Catatan Tambahan PTJ	**<u>Tugasan Pengajaran dan Pembelajaran (P&P)</u>** Antara kursus yang terlibat: 1. IEK 212/3 Process Heat Transfer 2. IEK 213/3 Mass Transfer and Separation Process 3. IEA 216/3 Computer Application in Industry 4. IBK 314 Downstream Process Technology 5. IBA407 Practical in Downstream Processing 6. IBG307 Bioprocess Optimization And Simulation
19.	Perakuan Jawatankuasa Pemilih	-
20.	Nota Rujukan JKPTJSM	

LAMPIRAN E

2. PENGESAHAN DALAM JAWATAN BAGI STAF AKADEMIK

5-E	**NAMA**	**TS. DR. ANA MASARA BINTI AHMAD MOKHTAR**
1.	**No. Staf**	0362/20
2.	**No. Kad Pengenalan**	910928146334
3.	**Jawatan**	Pensyarah Universiti DS13
4.	**Tarikh Lantikan:**	1 April 2024
5.	**Bidang Pengkhususan:**	Biokimia
6.	**Scopus Citation Index**	6
7.	**Markah LPP (%): 3 tahun terkini**	Tahun: Tahun 1 (2024) / Tahun 2 (2025) / Purata Markah: 94.10 / 92.47 / 93.28
8.	**Jawatankuasa dimaklumkan bahawa beliau telah memenuhi syarat untuk pengesahan jawatan iaitu:**	
9.	Peperiksaan / Keputusan	▪ Penilaian Program Transformasi Minda — LULUS ▪ BM Peringkat Maju (Ujian Berkuliah) — LULUS ▪ BM Peringkat Maju (Ujian Lisan & Ujian Bertulis) — LULUS ▪ Kursus Sijil Pengajaran & Pembelajaran — LULUS
10.	Catatan panel JK Pemilih yang lepas	**Mesyuarat ke-269 Jawatankuasa Pemilih, Pengesahan, Kenaikan Pangkat dan Lanjutan Perkhidmatan Bagi Staf Kumpulan Pengurusan & Profesional (Akademik) pada 29Januari 2024. Minit disahkan pada 13n Februari 2024.** Jawatankuasa memperakukan pelantikan baharu Dr. Ana Masara Ahmad Mokhar sebagai Pensyarah Universiti DS51 (Tetap-Percubaan) berdasarkan justifikasi yang telah dinyatakan oleh Pusat Pengajian serta telah memenuhi syarat lantikan tetap. Pelantikan ini berkuat kuasa pada 1 April 2024 dengan gaji pokok sebanyak RM 6,535.00. **Mesyuarat Jawatankuasa Kecil PTJ pada 22 Januari 2024.** Justifikasi lantikan: Keperluan kepakaran dalam pembelajaran dan pengajaran. Dr. Ana Masara Ahmad Mokhtar merupakan tenaga pengajar beberapa kursus-kursus umum di Pusat Pengajian Teknologi Industri. Keperluan kepakaran dalam penyelidikan dan penyeliaan pelajar ijazah tinggi (Master dan Ph.D.) Beliau menyumbang kepada penyelidikan, penyeliaan dan penerbitan yang dapat membantu meningkatkan prestasi Bahagian, Pusat Pengajian dan Universiti.

5-E	NAMA	TS. DR. ANA MASARA BINTI AHMAD MOKHTAR
		Pengajaran: IBK402/4 Bioproduct Development IBG112/3 Bioanalysis I IBG211/3 Bioanalysis II IBG207/3 Cell and Tissue Culture Technology IUK208/3 Experimental Design with Computer Applications IBG205/3 DNA and Metabolite Technology IBA404/8 Bioprocess Technology Research Project Penyeliaan: 6 pelajar pascasiswazah (Sarjana) (Penyelia Utama) 4 pelajar pascasiswazah (Sarjana) (Penyelia bersama) Penyelidikan Geran Antarabangsa ICGEB Geran FRGS pada tahun 2023 Geran Universiti Penerbitan: Beliau aktif dalam penerbitan iaitu 8 artikel (3 artikel Q1/Q2) dan 7 Chapter in Book pada tahun 2022 dan 2023. <u>Perakuan:</u> Berdasarkan justifikasi lantikan yang tersenarai di atas dan calon berdedikasi serta mempunyai aspirasi yang tinggi yang dapat dimanfaatkan oleh pusat pengajian dan universiti, Jawatankuasa memperakukan pelantikan semula staf akademik (Warganegara) Dr. Ana Masara Ahmad Mokhar bagi tempoh dua (2) tahun berkuat kuasa pada 1 April 2024 hingga 31 Mac 2026. Jawatankuasa memperakukan pelantikan tetap kepada Dr. Ana Masara Ahmad kerana beliau telah memenuhi syarat untuk lantikan tetap berdasarkan Hebahan Pendaftar Bil. 9 Tahun 2024 – Penetapan Syarat Lantikan Tetap Dan Syarat Pengesahan Jawatan Staf Akademik.
11.	Perkhidmatan lepas dengan institusi / agensi awam	Tiada USM: Kontrak 1 : 03.08.2020 hingga 02.02.2021 (1 tahun) – DS45 Kontrak 2 : 03.02.2021 hingga 01.04.2021 (2 Bulan) – DS45 Kontrak 3 : 03.05.2021 hingga 01.04.2022 (1 tahun) – DS45 Kontrak 4 : 01.04.2022 hingga 31.03.2024 (2 tahun) – DS51
12.	MAKLUMAT TAMBAHAN Senarai Pencapaian Penyelidikan dan Penerbitan (M1)	**Penerbitan** **A. Corresponding Author (2024-2025)** **Published** 1. Balasubramaniam, M., & Mokhtar, A. M. A. (2024). Past and present discovery of the BAFF/APRIL system – A bibliometric study from 1999 to 2023. Cellular Signalling, 120, 111201. <u>https://doi.org/10.1016/j.cellsig.2024.111201</u> 2. Balasubramaniam, M., Sapuan, S., Hashim, I. F., Ismail, N. I., Yaakop, A. S., Kamaruzaman, N. A., & Ahmad Mokhtar, A. M. (2024). The properties and mechanism of action of plant immunomodulators in regulation of immune response; 2013; A narrative review focusing on

5-E	NAMA	TS. DR. ANA MASARA BINTI AHMAD MOKHTAR
		Curcuma longa L., Panax ginseng C. A. Meyer and Moringa oleifera Lam. Heliyon, 10(7). https://doi.org/10.1016/j.heliyon.2024.e28261 3. Mohd Fadzil, F. N., Ngee, C., Alias, M., Lukman, M., Yaakop, A., Mohd Zaini Makhtar, M., & Ahmad Mokhtar, A. M. (2024). Application of Membrane-less Microbial Fuel Cell in Reducing Human Hazards from Dewatered Sludge. Pertanika Journal of Science and Technology, 32. https://doi.org/10.47836/pjst.32.3.12 **Under review / submitted** 1. Cdc42 Cys81Tyr Mutation Links Structural Destabilization to Malignant and Inflammatory Pathogenesis 2. Biosensors for Detecting Small RhoGTPases: Monitoring Expression and Activation 3. Impact of Flavonoids on Rho GTPase Mediated Cancer Signalling Pathway. 4. Assessing the Effect of Microbial Fuel Cell-Treated Dewatered Sludge Against Human Inflammatory Axis: In silico and In Vitro Analyses 5. Investigating the Kinases in Control of RhoH and the Impact of Aberrant RhoH Expression on Different Cytokines Production: A Systematic Review 6. A Bibliometric analysis: The Overview of RhoGTPases signalling in Cancer from 1990 to 2024 7. A systematic review of the characteristics of RhoH abnormalities in human diseases between 2000-2022 8. RhoH Y38X Mutation And Its Oncogenic Potential In Promoting Primary Immunodeficiency Malignancies 9. A Systematic Literature Review on the Evaluation of Microbial Fuel Cell (MFC) in Bioremediating Toxic Compounds from Wastewater Sludge **Co-author (2024-2025)** 1. Ng, Z.J., Goh, C.F., Mokhtar, A.M.A. et al. Microemulsion-based formulation of enterocin CC2: a novel antimicrobial solution targeting Streptococcus mutans. Drug Deliv. and Transl. Res. (2025). https://doi.org/10.1007/s13346-025-01823-x 2. Chan, W., Che Zain, M. & Ahmad Mokhtar, A. Microencapsulation of Elaeis guineensis leaf extract powder enriched with bioflavonoid preserves its antioxidant and anti-inflammatory properties: spray drying optimization and powder characterization. Food Measure (2024). https://doi.org/10.1007/s11694-024-02960-6 3. Sathasivan, D., Mohideen, M., Hafiidz Jaafar, M., Masara, A., Mokhtar, A., & Kamaruzaman, N. A. (n.d.). Asia-Pacific Journal of Science and Technology https://www.tci-thaijo.org/index.php/APST/index The antimicrobial resistance (AMR) scenario in Malaysia: A review of current trends, perspectives and strategic action. https://www.tci-thaijo.org/index.php/APST/index 4. Saw, K. C., Ahmad Mokhtar, A. M., & Ismail, N. I. (2024). Identification of potential SARS-CoV-2 Nsp1 inhibitors from Piper sarmentosum Roxb. using molecular docking. Tropical biomedicine, 41(2), 166–175. https://doi.org/10.47665/tb.41.2.006 5. Ng, Z. J., Ng, H. S., Kee, P. E., Gan, C.-Y., Mokhtar, A. M. A., Zarin, M. A., Lee, C. K., Tan, J. S., & Lan, J. C.-W. (2024). Optimization of ionic liquid-based aqueous two-phase system for partial purification of bacteriocin-like inhibitory substance from Enterococcus faecium CC2

5-E	NAMA	TS. DR. ANA MASARA BINTI AHMAD MOKHTAR
		and its inhibitory activity against Streptococcus mutans. Journal of the Taiwan Institute of Chemical Engineers, 160, 105347. https://doi.org/https://doi.org/10.1016/j.jtice.2024.105347 6. Ayyal Salman, H., Yaakop, A. S., Al-Rimawi, F., Ahmad Makhtar, A. M., Mousa, M., Semreen, M. H., & Alharbi, N. S. (2024). Ephedra alte extracts' GC-MS profiles and antimicrobial activity against multidrug-resistant pathogens (MRSA). Heliyon, 10(5), e27051. https://doi.org/10.1016/j.heliyon.2024.e27051 7. Sreedharan, D., Alias, H., Makhtar, M., Shun, T., Mokhtar, A., Shukor, H., Siddiqui, M., Alam, M., Kapoor, R. & Rafatullah, M. (2024). Screening of different growth conditions of Bacillus subtilis isolated from membrane-less microbial fuel cell toward antimicrobial activity profiling. Open Life Sciences, 19(1), 20220809. https://doi.org/10.1515/biol-2022-0809 **B. Geran Penyelidikan (Aktif Sahaja)** 1. Short-term Grant 2- Unveiling the Role of RhoH in Regulating NK Cell-Mediated Cytotoxicity on Non-Hodgkin B-Cell Lymphoma, Universiti Sains Malaysia (RM 38,000, 2025-2027) 2. Fundamental Research Grant Scheme - Elucidate cancer progression caused by faulty T cell receptor (TCR) signalling in Cdc42 C81Y-induced primary immunodeficiencies, National (RM 191100, 2022-2025) 3. International Centre for Genetic Engineering and Biotechnology (ICGEB) grant- To elucidate the effect of RhoH Y38X mutation in regulating Rac-induced BAFF-R signalling in PID, International (EURO 30,000, 2022-2025)
13.	Senarai Penglibatan Pengajaran dan Penyeliaan (M2)	**A. Kursus Pengajaran** 1. IBG112: Bionalisis I 2. IBG211: Bioanalisis II 3. IBG205: Teknlogi DNA dan Metabolit 4. IBG207: Teknologi Kultur Sel dan Tisu 5. IBK314: Teknologi Proses Hiliran 6. IBK317: Matematik Fisiologi 7. IBK318: Farmakogenomik 8. IBK319: Rekabentuk dan Pembangunan Bioproduk 9. IBA306: Projek Penyelidikan Teknologi Bioproses **B. Penyeliaan projek tahun akhir** 1. Nur Nabila Natasha Binti Arifin, Bachelor of Technology (Bioprocess), Investigating the potential of Kombucha in regulating Cdc42 activation status, Main supervisor, Ongoing (2025- current) 2. Lee Ke Lin, Bachelor of Technology (Bioprocess), Investigating the Role of Biosilica Derived from Cultivated Algae in Regulating Inflammaging, Main supervisor, Ongoing (2025- current) 3. Hasifah Binti Osman, Bachelor of Technology (Bioprocess), Investigating the Effect of Glu171Lys RhoG Mutation in Promoting

5-E	NAMA	TS. DR. ANA MASARA BINTI AHMAD MOKHTAR
		Cancer in Primary Immunodeficiency, Main supervisor, Ongoing (2025- current) 4. Nurshahira Binti Mohamad Noor, Bachelor of Technology (Bioprocess), Investigating Anti-Inflammatory Properties of Bacteriocin - Like Protein from *B. Subtilis, L. Plantarum*, and *L. Paracasei*, Main supervisor, Graduated (2022- 2023) 5. Mohammed Zharif Asyrani Bin Mohammed Alias, Bachelor of Technology (Bioprocess), Identifying the effect of membraneless microbial fuel cell (ML-MFC)- treated dewatered sludge on cytoxicity and cancer progression, Main supervisor, Graduated (2021- 2022) 6. Chen Sep Ngee, Bachelor of Technology (Bioprocess), Investigating the effect of membrane-less microbial fuel cell (ML-MFC)-treated dewatered sludge on inflammation, Main supervisor, Graduated (2021- 2022) 7. Nur Liyana Athilah Bt Mohd Affandi, Bachelor of Technology (Bioprocess), A comparative analysis of anti-inflammatory, anti-hyperglycemic and cytotoxic properties of *Streblus asper* and *Christia vespertilionis* aqueous leaves extracts, Main supervisor, Graduated (2020- 2021) **C. Penyeliaan Pelajar Ijazah Tinggi (aktif)** 1. Ahmed Ayesha Fatima, MSc in Bioprocess Technology, Investigating the Therapeutic Potential of Cocoa Waste Flavonoids in Regulating CDC42 Activation, Main Supervisor (2024-current) 2. Chong Chien Fung, MSc in Bioprocess Technology, Elucidating the Impact of the Cdc42 Cys81Tyr Mutation on Activation Status and Cancer-Associated Processess, Main Supervisor (2023-current) 3. Nurul Syafawani Binti Fizal, MSc in Bioprocess Technology, To Elucidate the Effect of RHOH Y38X Mutation in Regulating Rac-Induced Baff-R Signalling in Pid Malignancy, Main Supervisor (2023-current) 4. Nik Yasmin Umaira Binti Hasnizan, MSc in Bioprocess Technology, Elucidating the Effect of Cdc42 C81Y-Induced Primary Immunodeficiency in Promoting Cancer-Related Pathways in T-Cell, Main Supervisor (2023-current) 5. Saw Keat Chuan, MSc in Bioprocess Technology, Investigating the Role of the RhoH Y38X Mutation in Modulating Cancer-Related Cytokine Release, Main Supervisor (2023-current) 6. Fatin Nur Izzati binti Mohd Fadzil, MSc in Bioprocess Technology, Assessing the Effect of Microbial Fuel Cell (MFC)-Treated Dewatered Sludge Against Human Inflammatory Axis: *In Silico* and *In Vitro* Analyses, Main Supervisor (2022-current) 7. Muggunna A/P Balasubramaniam, MSc in Bioprocess Technology,

5-E	NAMA	TS. DR. ANA MASARA BINTI AHMAD MOKHTAR
		The Immunomodulatory Effect of *Christia Vespertilions* Leaf Extract (CVLE) on Baff-Mediated B Cell Proliferation and Differentiation, Main Supervisor (2022-current) 8. Eugene Ng Ee Chen, MSc in Biotechnology, School of Biological Sciences, Co-Supervisor (2024-current) 9. Nalle Trowce Lista, PhD in Pharmacy, National Poison Centre, Co-Supervisor (2023-current) 10. Arash Syafiq Bin Anuar, MSc in Biotechnology, School of Biological Sciences, Co-Supervisor (2023-current) 11. Jannarthaanah A/L Murugiah, MSc in Biotechnology, School of Biological Sciences, Co-Supervisor (2023-current) 12. Muhammad Adib Fadhlullah Bin Muhammad Lukman, MSc in Pharmacy, National Poison Centre, Co-Supervisor (2023-current) 13. Linesh Kumar A/L Selvaraja, MSc in Bioprocess Technology, School of Industrial Technology, Co-supervisor, (2021-Current) **D. Penyeliaan Pelajar Ijazah Tinggi (telah bergraduasi)** 1. Teh Jia Lin, MSc in Biotechnology,School of Biological Sciences, Co-supervisor, Graduated 2023
14.	Senarai Pencapaian Sanjungan Akademik dan Anugerah (M3)	**2024-2025** 1. Young Scientist 74th Lindau Nobel Laureate Meeting (Chemistry), (2025) 2. Early Career Member Travel award from Biochemcal society, UK (2025) 3. Member of Young Scientists Network- Academy of Sciences Malaysia (2024) 4. Professional Technologist, MBOT (2024) 5. Associate Editor, Probiotic and Antimicrobial Protein (2024)
15.	Senarai Penglibatan Konsultansi (M4)	**Konsultansi 2024-205** 1. R&D Consultant TRNCO Greentech Sdn Bhd (2025) 2. R&D Consultant, Algae Pure Genius Sdn Bhd (2024) 3. Anti-inflammatory testing, Vitacare (2024) 4. Anti-Inflammatory properties of EO, Gritz Sdn Bhd (2024)
16.	Senarai Penglibatan Sumbangan Kepada Universiti dan Komuniti (M5)	**Penglibatan sumbangan kepada Universiti dan komuniti** **A. Komuniti** 1. Scientific Committee FAOBMB 2027, International (2025-2027) 2. NSC 2025 Organising Committee – State Level Subcommittee Members, National (2025) 3. AJK Projek YSN-ASM Membership Drill, National (2025) 4. Juri STERF MRSM Kepala Batas, State (2024-2025) 5. Project leader, Global Ambassador Program - Promoting antimicrobial resistance awareness globally, SEOHUN (2024)

5-E	NAMA	TS. DR. ANA MASARA BINTI AHMAD MOKHTAR
		6. Project leader, AMbassadoR program- Promoting Antimicrobial Resistance Awareness among Matriculation college students, BJIM USM (2024) 7. Juri MARA STERF, National (2024) 8. Speaker for Alt Protein Project @ MRSM Kepala Batas, State (2024) 9. Panel for MOCK interview, MRSM Kepala Batas, State (2024) 10. Jury, Product innovation, Institut Kimia Malaysia, State (2024) 11. Speaker and panel for Malaysian International Food and Beverage Trade Fair Malaysian, International Food and Beverage Trade Fair, International (2024) 12. Speaker Wildlife destruction cases in Malaysia, Tech Dome, State (2024) 13. Panel Environmental Impact Challenge, Tech Dome, State (2024) 14. Speaker & Mentor for Schoolwide Enrichment Model (SEM) Programme, MRSM Kepala Batas, State (2024) 15. Speaker for STEM, MRSM Sungai Besar, National (2024) 16. Speaker for STERF MRSM Transkrian, State (2024) 17. Penyelia HFF (2024) **B. Universiti/ PTJ** 1. AJK Kualiti Program Akademik Mod Penyelidikan Pascasiswazah PPTI, School, (2025-2026) 2. AJK Pemasaran dan Penjenamaan Program PPTI, School, (2025-2026) 3. MyRA secretariat PPTI, School (2024) 4. Penyelia HFF225 (2024) 5. Panel penilai Pembentangan Cadangan Penyelidikan, School (2024) 6. JKPD- Committee member, School (2022 – 2024) 7. OBE & PPTI accreditation- Committee member, School (2022 – 2024)
17.	Kualiti Pengajaran (maklum balas pelajar)	**2024-2025** 1. IBG112: Bionalisis I: 98.4% Amat setuju 2. IBG211: Bioanalisis II: 99.4 -100 % Amat setuju 3. IBG205: Teknlogi DNA dan Metabolit: 99.0% Amat setuju 4. IBG207: Teknologi Kultur Sel dan Tisu: 99.7% Amat setuju 5. IBK314: Teknologi Proses Hiliran: 100% Amat setuju 6. IBK317: Matematik Fisiologi: 100% Amat setuju
18.	Penilaian KJ tentang pengajaran dan pembelajaran	Lihat perkara no. 20

5-E	NAMA	TS. DR. ANA MASARA BINTI AHMAD MOKHTAR

19. Syarat Pengesahan

A. Syarat Umum Pengesahan Jawatan Staf Akademik

SYARAT UMUM PENGESAHAN JAWATAN		PENCAPAIAN
1.	Penilaian Program Transformasi Minda	Lulus
2.	Lulus Modul Pengajaran & Pembelajaran anjuran Pusat Pembangunan Kecemerlangan Akademik (CDAE) Penilaian Ceraoan Pengajaran Penilaian E-Portfolio Pengajaran	Lulus
3.	Lulus Modul Penyeliaan berstruktur anjuran Pusat Pembangunan Kecemerlangan Akademik (CDAE)	Lulus
4.	Lulus Bahasa Malaysia Peringkat Maju (Ujian Lisan, Ujian Bertulis dan Ujian Penyampaian Kuliah)	
	- Ujian Bertulis dan Lisan	Lulus
	- Ujian Berkuliah	Lulus
5.	Menyempurnakan pengajaran sekurang-kurangnya dua (2) semester dengan melengkapkan semua dokumentasi kursus-kursus yang terlibat	√
6.	Diperakukan oleh Ketua Jabatan	√
7.	Markah LPP berdasarkan purata melebihi : a.Tahun 1 : 85% b.Purata Tahun 1 dan Tahun 2 : 80% c.Purata Tahun 1, Tahun 2 dan Tahun 3: 80 %	√

B. Syarat PTJ Bidang Kejuruteraan dan Teknologi

BIL	SYARAT	PENCAPAIAN
1.	**Menghasilkan penerbitan (WoS/Scopus / Buku)** **Tiga (3) Penerbitan sebagai Corresponding Author / First Author dalam WoS;** **ATAU** Satu (1) artikel dalam Jurnal berindeks WoS sebagai Corresponding Author/First Author **DAN** dua (2) bab dalam buku penyelidikan berindeks WoS/Scopus/MAPIM; **ATAU** Satu (1) artikel dalam Jurnal berindeks WoS sebagai Corresponding Author/First Author **DAN** satu (1) buku penyelidikan berindeks WoS/Scopus/MyCITE. Nota: Penerbitan adalah dalam bidang kepakaran staf akademik.	3 penerbitan dalam WOS (Rujuk Senarai 1)
2.	**Menyelia pelajar di peringkat Master / PhD** **Menyelia satu (1) pelajar PhD sebagai Penyelia Bersama (sekurang-kurangnya 12 bulan)**	Penyelia bersama 1 Pelajar siswazah PHD

5-E	NAMA	TS. DR. ANA MASARA BINTI AHMAD MOKHTAR

	DAN Telah menggraduatkan dua (2) pelajar siswazah Mod Campuran/Kerja kursus MSc sebagai Penyelia Utama **ATAU** **Menyelia satu (1) pelajar siswazah MSc/PhD Mod Penyelidikan sebagai Penyelia Utama (sekurang-kurangnya enam (6) bulan bagi MSc dan 12 bulan bagi PhD)** **ATAU** Menyelia dua (2) pelajar Mod Campuran sebagai Penyelia Utama.	Penyelia utama 7 Pelajar siswazah Msc Mod Penyelidikan Penyelia Bersama 7 Pelajar siswazah Msc Mod Penyelidikan (Rujuk Senarai 2)
3.	**Geran penyelidikan** **Telah mendapat Geran dalaman sebagai Penyelidik Utama.** **DAN** **Telah memohon satu (1) Geran luar (Kebangsaan/Industri/Antarabangsa) sebagai Penyelidik Utama**.	3 geran aktif sebagai penyelidik utama 1 Permohonan geran Wellcome trust (Rujuk Senarai 3)
4.	**Lain-lain syarat** Menghantar borang permohonan jurutera professional **DAN** **Khidmat atau sumbangan kepada PTJ atau Universiti.**	Professional Technologist (MBOT) 3 copyrights (Rujuk Senarai 4)

Senarai 1

1. Balasubramaniam, M., & Mokhtar, A. M. A. (2024). Past and present discovery of the BAFF/APRIL system – A bibliometric study from 1999 to 2023. Cellular Signalling, 120, 111201. https://doi.org/10.1016/j.cellsig.2024.111201
2. Balasubramaniam, M., Sapuan, S., Hashim, I. F., Ismail, N. I., Yaakop, A. S., Kamaruzaman, N. A., & Ahmad Mokhtar, A. M. (2024). The properties and mechanism of action of plant immunomodulators in regulation of immune response; 2013; A narrative review focusing on Curcuma longa L., Panax ginseng C. A. Meyer and Moringa oleifera Lam. Heliyon, 10(7). https://doi.org/10.1016/j.heliyon.2024.e28261
3. Mohd Fadzil, F. N., Ngee, C., Alias, M., Lukman, M., Yaakop, A., Mohd Zaini Makhtar, M., & Ahmad Mokhtar, A. M. (2024). Application of Membrane-less Microbial Fuel Cell in Reducing Human Hazards from Dewatered Sludge. Pertanika Journal of Science and Technology, 32. https://doi.org/10.47836/pjst.32.3.12

Senarai 2

Penyelia Utama
1. Ahmed Ayesha Fatima
2. Chong Chien Fung
3. Nurul Syafawani Binti Fizal
4. Nik Yasmin Umaira Binti Hasnizan
5. Saw Keat Chuan
6. Fatin Nur Izzati binti Mohd Fadzil
7. Muggunna A/P Balasubramaniam

Penyelia Bersama
1. Nalle Trowce Lista, PhD in Pharmacy,
2. Eugene Ng Ee Chen, MSc in Biotechnology
3. Arash Syafiq Bin Anuar, MSc in Biotechnology,

5-E	NAMA	TS. DR. ANA MASARA BINTI AHMAD MOKHTAR
	4. Jannarthaanah A/L Murugiah, MSc in Biotechnology, 5. Muhammad Adib Fadhlullah Bin Muhammad Lukman, MSc in Pharmacy 6. Linesh Kumar A/L Selvaraja, MSc in Bioprocess Technology 7. Teh Jia Lin, MSc in Biotechnology (Graduated) **<u>Senarai 3</u>** Active 1. Short-term Grant 2, Universiti Sains Malaysia (University), Unveiling the Role of RhoH in Regulating NK Cell-Mediated Cytotoxicity on Non-Hodgkin B-Cell Lymphoma, RM 38,000, 2 years (1 July 2025 – 30 June 2027) 2. Fundamental Research Grant Scheme (National), Elucidate cancer progression caused by faulty T cell receptor (TCR) signalling in Cdc42 C81Y-induced primary immunodeficiencies, RM 191,100, 3 years, (Sept 2022- August 2025) 3. International Centre for Genetic Engineering and Biotechnology (ICGEB) grant (International), To elucidate the effect of RhoH Y38X mutation in regulating Rac-induced BAFF-R signalling in PID, EURO 30,000, 4 years, (Jan 2022- Dec 2025) Applied 1. Career Development Award, Rho-ttacking Non-Hodgkin B-Cell Lymphoma: Decoding the Role of Small Rho GTPases in Regulating NK Cell-Mediated Cytotoxicity- Unsuccessful 2. Eric Reid Fund for Methodology grant, Biochemical Society UK- Unsuccessful **<u>Senarai 4</u>** 1. Ana Masara Binti Ahmad Mokhtar, Chong Chien Fung, Nik Yasmin Umaira Binti Hasnizan, Nurul Syafawani Binti Fizal, Saw Keat Chuan, Recapitulating Protein Mutations In Vitro, LY2025P00582, 05/02/2025, LITERARY, Registered (MyIPO) 2. Ana Masara Binti Ahmad Mokhtar, Fatin Nur Izzati, Binti Mohd Fadzil, Muhammad Adib Fadhlullah Bin Muhammad Lukman, Nur Azzalia Kamaruzaman, RISK Evaluation Of Dewatered Sludge Impact On Human Health And Safety, LY2025P00583, 05/02/2025, LITERARY, Registered (MyIPO) 3. Ana Masara Binti Ahmad Mokhtar, Amira Suriaty Binti Yaakop, Nor Asniza Binti Ishak, Nor Hawani Binti Salikin, Noreen Suliani Binti Mat Nanyan, Nurul Izza Binti Ismail, Syafinaz Binti Amin Nordin, AMBASSADOR: Antimicrobial Resistance (Amr) Awareness Module, LY2025P00581, 05/02/2025, LITERARY, Registered (MyIPO)	
20.	Perakuan Ketua Jabatan	1. Kebolehan menggunakan Bahasa Malaysia: Sangat fasih menggunakan Bahasa Malaysia untuk komunikasi, pendengaran dan penulisan. 2. Tugas-Tugas Pengajaran: Dapat melakukan tugasan pengajaran dalam kelas dan amali dengan sempurna. Gaya penyampaian dalam kelas dapat menarik perhatian pelajar. 7 Subjek UG mengajar: IBG207, IBG 112, IBG 211, IBK317, IBK318, IBK314, IBG319. 3. Penyelidikan dan Penerbitan - Ts. Dr. Ana Masara aktif dalam bidang penyelidikan dan mempunyai 3 geran aktif yang berjumlah RM RM 374,100.00. Penerbitan journal sebagai coresponding author sejak tahun 2021 berjumlah 9. 4. Perkhidmatan kepada Universiti - Ts. Dr. Ana Masara terlibat dalam aktiviti seperti Pengacara 3 minute thesis, Pegawai/AJK Myra, dan Pegawai/AJK Kelestarian USM. 5. Perkhidmatan kepada Masyarakat - Ts. Dr. Ana Masara melibatkan diri dalam pelbagai perkhidmatan kepada masyarakat seperti Juri pertandingan inovasi STERV di MRSM, Speaker tentang SDG di peringkat daerah, SBP, sekolah, MARA, dan sebagai Global Ambassador Anti-microbial resistance (AMR) Program, USM-Prince Songkhla University. 6. Lain-lain - Baru-baru ini Ts Dr Ana Masara menerima pengiktirafan peringkat antarabangsa seperti 'Young Scientist 74th Lindau Nobel Laureate Meeting (Chemistry), Lindau Nobel Laureate Meeting'.	

5-E	NAMA	TS. DR. ANA MASARA BINTI AHMAD MOKHTAR
21.	Perakuan Jawatankuasa Kecil PTJ	Jawatankuasa Kecil PTJ – Sumber Manusia setelah meneliti prestasi dan menemuduga calon, bersetuju memperakukan Ts. Dr. Ana Masara Binti Ahmad Mokhtar disahkan dalam jawatan sebagai Pensyarah Universiti DS13.
22.	Perakuan Jawatankuasa Pemilih	

LAMPIRAN F

3. PELANTIKAN SEMULA STAF AKADEMIK KONTRAK WARGANEGARA PENSYARAH UNIVERSITI DS13 BAGI TUJUAN MEMENUHI SYARAT LANTIKAN TETAP

6-F	**NAMA**	**DR. SUMARNI BINTI MANSUR**
1.	Umur	35 tahun
2.	Warganegara	Malaysia
3.	Jabatan	Pusat Pengajian Teknologi Industri
4.	Kelayakan Akademik	Doktor Falsafah Universiti Teknologi Malaysia (UTM), Johor, Malaysia 2020 *Sarjana Falsafah* Universiti Teknologi Malaysia (UTM), Johor, Malaysia 2016 *Sarjana Muda Sains Bioperubatan (Kepujian)* Universiti Islam Antarabangsa (UIA), Malaysia 20014
5.	**H-Index** **1. Scopus** **2. WoS**	 Scopus – 11 WoS – 12
6.	Bidang Kepakaran	Membrane technology, Functional materials, Biomass, Hemodialysis membrane, Biocompatible materials
7.	Lantikan Lepas	Sedia Ada (Kontrak I): 23.11.2023 hingga 22.11.2025 (2 tahun)
8.	**<u>Sasaran KPI</u>** yang ditetapkan oleh PTJ tahun lantikan semasa (Maklumat sasaran KPI untuk tujuan pelantikan. *Nota: Perakuan Mesyuarat JKPTJSM ke-84 bertarikh 10 Oktober 2023. KPI terkini 2025 adalah selaras dengan Hebahan Pendaftar Bil. 11/2025 – Penetapan sasaran minimum tahunan pensyarah universiti dan proses pelantikan staf akademik baharu bertarikh 17 Februari 2025.	**<u>SASARAN KPI LANTIKAN SEMASA 06.01.2023 – 05.01.2025</u>** **Pengajaran Prasiswazah dan Pascasiswazah** Minimum 6-unit untuk 1 sidang. Penyeliaan 3-5 pelajar FYP dan 3-5 Latihan Industri per sidang **Penerbitan** Q1/Q2 WoS – 4 setahun. Book chapters setiap tahun **Pelajar Pascasiswazah** Minimum 1 pelajar pascasiswazah mod penyelidikan setahun. **Geran Penyelidikan** 1 geran kebangsaan/industri/antarabangsa setahun. **Penjanaan Kewangan** Minimum 2 aktiviti penjanaan kewangan seperti konsultasi, kursus-kursus profesional dan lain-lain.

6-F	NAMA	DR. SUMARNI BINTI MANSUR

9. **PENCAPAIAN** KPI yang ditetapkan oleh PTJ (Tahun Lantikan semasa)

PENCAPAIAN LANTIKAN SEMASA

1. PENERBITAN

Sasaran / Pencapaian	Artikel (WoS)/ Scopus ❖ Q1/Q2 WoS	Bab Dalam Buku
2024	2 published ▪ 2 co-author/sole-author	1 published
2025	5 published ▪ 2 co-author/sole-author ▪ **3 main author**	1 under review

A. JOURNAL (MAIN AUTHOR/CORRESPONDING)
Status: 3 published, 2 under review

B. BAB DALAM BUKU
Status: 1 published, 1 under review

2. PENYELIDIKAN - GERAN BARU

Tahun	Geran Penyelidikan
2024	▪ Mempunyai 1 geran yang aktif (2024) - 1 Geran Jangka Pendek USM

3. PENYELIAAN PELAJAR SISWAZAH

Tahun/Status	Pelajar Master	Pelajar PhD
2024	-	-
2025	1	-

4. SUMBANGAN KEWANGAN KEPADA PPTI

Tahun	Aktiviti	Jumlah sumbangan (RM)
2024-2025	-	-

10. Senarai Pencapaian Penyelidikan dan Penerbitan (M1)

(A) Senarai jurnal (published)

2024

1. **Co-author**, Peechmani P., Makhtar S.N.N.M., **Sumarni Binti Mansur**, Othman M.H.D., Kamaludin R., Puteh M.H., Wong K.Y., Kurniawan T.A., Rahman M.A., Jaafar J., Ismail A.F., 2024, Uncovering the potential of yttrium doped zinc oxide nanoparticles as an efficient catalyst for photodegradation of recalcitrant pollutant and a contributor antibacterial property. JOURNAL OF WATER PROCESS ENGINEERING), Impact Factor: 6.3, DOI:10.1016/j.jwpe.2023.104706, WoS
2. **Co-author**, Iftikhar, Musawira, Othman, Mohd Hafiz Dzarfan, Khan, Imran Ullah, Ismail, Nurul Jannah, Imtiaz, Aniqa, Kamaludin, Roziana, Kadir, Siti Hamimah Sheikh Abdul, Puteh, Mohd Hafiz, Abdullah, Huda, **Sumarni Binti Mansur**, 2024, Future research perspectives in hemodialysis membrane technology, JOURNAL

6-F	NAMA	DR. SUMARNI BINTI MANSUR
		OF INDUSTRIAL AND ENGINEERING CHEMISTRY, 138,72-103, Impact Factor: 5.9 DOI:10.1016/j.jiec.2024.04.024, WoS **2025** 1. **Co-author**, Johari, A.F., Zainol Abidin, M.N., Mohamad, S., Mohamad, S.F., **Mansur, S**., 2025, Surface Modification of Polyimide Membrane Via Radiation Induced Graft Copolymerization of Methyl Methacrylate: Effects of Monomer Concentration and Radiation Dose. Journal Defect and Diffusion Forum, 440, 29-35, Scopus 2. **First-author**, **Mansur, S**., Wang, H., Lee, C. K., Usman, S. N., Haikang, H., Hu, C., Zeng, G. Tay., G. S., 2025, Utilization of Anhydrides and Monoglycerides for Ultra-Violet Curable Alkyd Resin Preparation via Solvent-Free Enzymatic Process, Journal of Applied Polymer Science, WoS 3. **First-author**, **Mansur, S**., Wang, H., Lee, C. K., Usman, S. N., Haikang, H., Hu, C., Zeng, G. Tay., G. S., 2025, Enzymatic synthesis of Ultraviolet curable alkyd resin and modification with succinic anhydride, Journal of Applied Polymer Science, WoS 4. **First-author and corresponding-author**, **Sumarni Mansur**, Noresah Said, Muhammad Nidzhom Zainol Abidin, Ahmad Fauzi Ismail, 2025, Development of antifouling membranes incorporating graphene oxide-iron oxide nanocomposites for potential Hemodialysis applications, Journal of Porous Materials, WoS 5. **Co-author,** Nasir A.H.M.P., Norizan M.N., Saharudin N.I., Mansur S., Knight V.F., Norrahim M.N.F., (2025). Plastic waste carbonization - a review. Part 1. Morphology and characteristics of carbonization products, Polimery Polymers, 70(5) **(B) Senarai bab dalam buku (published)** 1. S. Mansur & M.N.Z. Abidin. (2024). 4 – Sputtering technique. In M.H.D. Othman, M.A. Rahman, T. Matsuura, M.R. Adam, & S.N.N.M. Makhtar (Eds.), Advanced Ceramics for Photocatalytic Membranes (pp. 75-99). Elsevier. https://doi.org/10.1016/B978-0-323-95418- 1.00001-X **(C) Senarai geran (aktif)** 1. **Sumarni Binti Mansur**, Nurul Fazita Binti Mohammad Rawi,Abdul Khalil Bin H. P. Shawkataly,Dr. Muhammad Nidzhom Bin Zainol Abidin, Elucidating the enhancement of physicochemical and electrochemical properties of modified geopolymer ion exchange membranes through the addition of biomass-derived ashes, SHORT TERM - ACADEMIC STAFF, RM 46,000.00, 01/08/2024 - 01/08/2026, (Active) 2. Mohd Ridhwan Bin Adam, Abdul Latif Bin Ahmad, Prof. Ts. Dr. Mohd Hafiz Dzarfan Othman, Low Siew Chun, **Sumarni Binti Mansur**, Mohd Faizal Bin Mohd Zulkifly, Development Of Composite Ceramic-Polymeric Membrane For Microplastics Remediation In Water, Transdisciplinary Research University Grant Scheme (Rutrans), Rm 119,900.00, 01/12/2024 - 30/11/2027, (Active)

6-F	NAMA	DR. SUMARNI BINTI MANSUR

(D) Senarai persidangan

1. Invited Speaker, Appointment As Invited Speaker For Regional Congress On Membrane Technology 2024 (RCOM2024), Impiana Hotel, Senai, Johor Bahru, Johor, Malaysia, Advanced Membrane Technology Research Center (AMTEC) and the Frontiers Material Research Alliance (FMRA) at Universiti Teknologi Malaysia (UTM), Regional Congress On Membrane Technology 2024 (RCOM2024) (International)
2. Invited Speaker, Appointment as invited speaker for 15th Conference of Aseania Membrane Society (AMS15), Berjaya Times Square, Kuala Lumpur, Malaysia.

11. Senarai Penglibatan Pengajaran dan Penyeliaan (M2)

(A). Subjek pengajaran (pelajar undergraduate)

No.	Year	Session/Semester	Subject
1	2024	2023/2024 (232)	IWK204/3, 232, 2024, Bioresource, Paper And Coatings Product Development , 34, 3
2	2024	2023/2024 (232)	IWK305/2, 232, 2024, Advanced Technology of Coatings, 40, 2
3	2024	2023/2024 (232)	IWK306/2, 232, 2024, Fibre and Lignocellulosic Composite, 59, 2
4	2023/2024	2023/2024 (236)	IWA313/8, 236, 2023, Research Project Of Bioresource, Paper And Coatings Technology , 34, 14
5	2024	2024/2025 (241)	IWA281/2, 241, 2024, Coatings Technology Laboratory I, 58, 2
6	2024	2024/2025 (241)	IWA383/2, 241, 2024, Paper Technology Laboratory Ii, 43, 2
7	2024	2024/2025 (241)	IWK201/4, 241, 2024, Raw Materials and Coatings Chemistry, 59, 3
8	2025	2024/2025 (242)	IWK204/3, 242, 2025, Bioresource, Paper and Coatings Product Development, 27, 2
9	2025	2024/2025 (242)	IWK305/2, 242, 2025, Advanced Technology of Coatings, 11, 1
10	2024/2025	2024/2025 (246)	IWA313/8, 246, 2024, Research Project of Bioresource, Paper and Coatings Technology, 42, 12

(B) Penyeliaan pelajar undergraduate (Projek Tahun Akhir)

1. Anishwary A/p Nagesh, 2024, Final Year Project, Synthesis and characterisation of wood ash-derived geopolymer for ion exchange process, Main Supervisor, Active
2. Nadhirah Binti Ahmad Hisham, 2024, Final Year Project, Study of ion-exchange property of rice husk ash-derived geopolymer, Main Supervisor, Active
3. Ali Hanafiah Muhammad, 2024, Final Year Project, Adsorptive removal of methylene blue dye by bamboo-derived graphene oxide, Main Supervisor, Active

6-F	NAMA	DR. SUMARNI BINTI MANSUR
		(C) Penyeliaan pelajar pasca siswazah (Master – Mode: Penyelidikan) **Penyelia Utama:** 1. Aidee Putera bin Kamal Suradee, 2025, Elucidating the enhancement of physicochemical and electrochemical properties of modified geopolymer ion exchange membranes through the addition of biomass-derived ashes.
12.	Senarai Pencapaian Sanjungan Akademik dan Anugerah (M3)	**2024** 1. Panel Juri Pesta Schoolwide Enrichment Model (SEM) Type III 2024, 2024, MRSM Kepala Batas (State) 2. Panel Mock Interview MRSM Kepala Batas, 2023, MRSM Kepala Batas (State) 3. Panel for Environmental Impact Challenge 2024, 2024, Tech Dome Penang, Penang Tech Centre (National) 4. Invited Speaker, Invitation to Speak at the Environmental Impact Challenge 2024, Tech Dome Penang, Komtar Geodesic Dome, Jalan Penang, 10000 Georgetown, Penang, Malaysia, Tech Dome Penang, Penang Tech Centre, Environmental Impact Challenge 2024 (State) 5. Invited Speaker, APPOINTMENT AS INVITED SPEAKER FOR REGIONAL CONGRESS ON MEMBRANE TECHNOLOGY 2024 (RCOM2024), Impiana Hotel, Senai, Johor Bahru, Johor, Malaysia, Advanced Membrane Technology Research Center (AMTEC) and the Frontiers Material Research Alliance (FMRA) at Universiti Teknologi Malaysia (UTM), REGIONAL CONGRESS ON MEMBRANE TECHNOLOGY 2024 (RCOM2024) (International) 6. Invited Speaker, JEMPUTAN SEBAGAI PENCERAMAH BAGI PROGRAM FESTIVAL OF IDEAS @ PPTI 1/2024, Bilik Persidangan, Pusat Pengajian Teknologi Industri, USM, Pusat Pengajian Teknologi Industri, USM, PROGRAM FESTIVAL OF IDEAS @ PPTI 1/2024 (State) 7. Members, Malaysia Membrane (MyMembrane) Society, 2024-2026 (National) 8. Invitation to Review Manuscript ID 3fb25075-57b2-4405-b630-5e347c1d6059, 2023, Environmental Processes 9. Invitation to Review Manuscript ID 61c543f1-09e8-4039-ab59-a35b4086923c, 2024, Cellulose 10. Invitation to Review Manuscript ID QAEN-2023-0123, 2024, Cogent Engineering **2025** 1. Invited Speaker, Appointment as invited speaker for 15th Conference of Aseania Membrane Society (AMS15), Berjaya Times Square, Kuala Lumpur, Malaysia. 2. Jemputan sebagai Juri Pesta Schoolwide Enrichment Model (SEM) TYPE III, MRSM Kepala Batas 3. Jemputan sebagain Penceramah Bengkel Design Thinking, MRSM Kepala Batas 4. Invitation to Review Manuscript: Optimization of Polypyyrole/Reduced -Graphene Oxide coated fabric via dip-coated and in-site polymerization as photothermal for solar-driven interfacial evaporation, 2025, MJFAS

6-F	NAMA	DR. SUMARNI BINTI MANSUR
		5. Invitation to Review Manuscript: Alternative raw materials for detergent industry, 2025, RSIS International Journals 6. Invitation to Review Manuscript: Review:Electrochemical Investigations of Al-8088/TiO2/SiC Hybrid Composites developed using stir Casting method. 2025, RSIS International Journals 7. Invitation to Review Manuscript: Review:A Review on "The Sustainable Scouring Processes of Organic Cotton Fabric", 2025, RSIS International Journals 8. Members, Malaysia Membrane (MyMembrane) Society, 2024-2026 (National)
13.	Senarai Penglibatan Konsultansi (M4)	Consultation Without Monetary Rewards (with appointment letter / consultation report) 1. Penceramah Bengkel "Membrane Manufacturing For Wastewater Treatment", Advanced Membrane Research Centre (AMTEC), Fakulti Kejuruteraan Kimia Dan Tenaga UTM Johor Bahru, Johor, 26/08/2024 - 27/08/2024, RM0.00 (University)
14.	Senarai Penglibatan Sumbangan Kepada Universiti dan Komuniti (M5)	**(A) <u>Sumbangan Kepada Universiti</u>** **<u>2024</u>** 1. School Committee,Setiausaha Jawatankuasa Sebutharga Bahagian Teknologi Biosumber, 02/10/2024-31/12/2024 (Department) 2. Programme Manager,Industrial Visit Coordinator, BIORESOUCE TECHNOLOGY INDUSTRIAL VISIT 2024, 01/09/2024-31/10/2024 (Department) 3. School Committee,Ahli Jawatankuasa Key Performance Indicator (KPI) PPTI, 01/07/2024-31/12/2024 (School) 4. School Committee,Ahli Jawatankuasa Jurnal Biomass Materials and Technology PPTI, 01/11/2023-30/10/2026 (School) 5. Programme Manager,Ketua Projek Jawatankuasa Virtual Career Talk with Experts Fellowship, 01/08/2024-17/08/2024 (School) 6. School Committee,PELANTIKAN SEBAGAI PENYELARAS SULAM BAHAGIAN TEKNOLOGI BIOSUMBER, 01/08/2024-31/12/2024 (School) 7. Others,PENGACARA MAJLIS BAGI PROGRAM KENDURIAN KELAB SUKAN DAN KEBAJIKAN PUSAT PENGAJIAN TEKNOLOGI INDUSTRI (KESAKTI), 21/08/2024-21/08/2024 (School) 8. Committee,AHLI JAWATANKUASA KELESTARIAN PUSAT PENGAJIAN TEKNOLOGI INDUSTRI, 22/03/2024-31/12/2024 (School) 9. Committee,AHLI JAWATANKUASA OBE & AKREDITASI PUSAT PENGAJIAN TEKNOLOGI INDUSTRI, 01/03/2024-31/12/2024 (School) 2025 1. Committee,AHLI JAWATANKUASA KELESTARIAN PUSAT PENGAJIAN TEKNOLOGI INDUSTRI, 15/1/2025-31/12/2026 (School) 2. Committee,AHLI JAWATANKUASA PEMBANGUNAN DAN PENGURUSAN PENYELIDIKAN, 3/3/2025-2/3/2026 (School)

6-F	NAMA	DR. SUMARNI BINTI MANSUR
		3. Committee,AHLI JAWATANKUASA OBE 7 AKREDITASI PUSAT PENGAJIAN TEKNOLOGI INDUSTRI, 15/1/2025-31/12/2026 (School) 4. Pelantikan sebagai penyelia kursus HFE224: Penghayatan Etika dan Peradaban, Semester 2, Sidang Akademik 2024/2025 **(B) <u>Sumbangan Kepada Komuniti</u>** 1. Volunteer, Invitation to Conduct Activities for Refugee Students on 1st August, Community, 01/08/2024-01/08/2024 (School)
15.	Maklumat Tambahan Calon (Sekiranya Ada)	-
16.	Maklumat Sasaran KPI Untuk Tujuan Pelantikan KPI terkini 2025 adalah selaras dengan Hebahan Pendaftar Bil. 11/2025 – Penetapan sasaran minimum tahunan pensyarah universiti dan proses pelantikan staf akademik baharu bertarikh 17 Februari 2025.	**1. M1: Penerbitan / Publications** Minimum dua (2) penerbitan setahun - penerbitan dalam jurnal berindeks sebagai *First Author/Corresponding Author* **2. M1: Penyelidikan / Research** 1 geran jangka pendek / kebangsaan/industri/antarabangsa setahun **3. M2: Pengajaran / *Teaching*** Minimum 8 unit per sidang akademik **4. M2: Penyeliaan / Supervision** a. Penyelian 3 - 5 pelajar FYP dan 3 - 5 Latihan Industri per sidang, b. 1 pelajar pasca siswazah (Sarjana) / c. 1 pelajar pasca siswazah (PhD) **5. Penjanaan Pendapatan / Income Generation** Minimum 2 aktiviti penjanaan kewangan seperti konsultasi, kursus-kursus profesional dan lain-lain
17.	Penetapan Gaji	Gaji Semasa Gred DS13 : RM6,636.60 Imbuhan Tetap Keraian : RM600.00 Imbuhan Tetap Perumahan : RM700.00 Bayaran Bantuan Sara Hidup : RM300.00 Kenaikan Gaji Tahunan : RM290.00 **Jumlah Keseluruhan : RM8,526.60**
18.	Berkhidmat di USM secara lantikan kontrak sekurang-kurangnya dua (2) tahun	Kontrak I: 23.11.2023 hingga 22.11.2025 (2 tahun)
19.	Markah LPP (%) Mencapai purata markah	Tahun: 2024, 2025, Purata Markah: 88.61, 89.66, 89.14

6-F	NAMA	DR. SUMARNI BINTI MANSUR

20.

No.	Title of Publication (author, title, journal, year, volume, pages) *First Author or Corresponding Author only*	Index (WoS/Scopus/My Cite/ERA/ Buku/Bab Dalam Buku)	First Author	Corresponding Author
1.	Sumarni Mansur, Noresah Said, Muhammad Nidzhom Zainol Abidin, Ahmad Fauzi Ismail, 2025, Development of antifouling membranes incorporating graphene oxide-iron oxide nanocomposites for potential Hemodialysis applications, Journal of Porous Materials, WoS	WoS	/	/
2.	Mansur, S., Wang, H., Lee, C. K., Usman, S. N., Haikang, H., Hu, C., Zeng, G. Tay., G. S., 2025, Enzymatic synthesis of Ultraviolet curable alkyd resin and modification with succinic anhydride, Journal of Applied Polymer Science, WoS	WoS	/	
3.	Mansur, S., Wang, H., Lee, C. K., Usman, S. N., Haikang, H., Hu, C., Zeng, G. Tay., G. S., 2025, Utilization of Anhydrides and Monoglycerides for Ultra-Violet Curable Alkyd Resin Preparation via Solvent-Free Enzymatic Process, Journal of Applied Polymer Science, WoS	WoS	/	
4.	Sumarni Binti Mansur, Abidin M.N.Z., 2024, Sputtering technique, in (Advanced Ceramics for Photocatalytic Membranes: Synthesis Methods, Characterization and Performance Analysis, and Applications in Water and Wastewater Treatment), N/A,75-99, (Published)	Bab Dalam Buku	/	

21.	Pengalaman pengajaran sekurang-kurangnya satu (1) semester	3 Semester - 2023/2024 (232) - 2024/2025 (241) - 2024/2025 (242)

22. Senarai Semak Syarat Lantikan Tetap

1.	Berkhidmat di USM secara lantikan kontrak sekurang-kurangnya dua (2) tahun pada 31.12.2023	√
2.	Mencapai purata markah Laporan Penilaian Prestasi (LPP) 85%	√
3.	Menerbitkan sekurang-kurangya empat (4) penerbitan dalam jurnal berindeks sebagai First Author/Corresponding Author ▪ Bidang Sains & Teknologi – WoS/Scopus ▪ Bidang Sastera - (WoS/Scopus/MyCite/ERA/Buku/Bab Dalam Buku)	3/4
4.	Pengalaman pengajaran sekurang-kurangnya satu (1) semester	√

23. Perakuan Jawatankuasa Pemilih Untuk Lantikan Lepas

Mesyuarat ke-265 Jawatankuasa Pemilih, Pengesahan, Kenaikan Pangkat dan Lanjutan Perkhidmatan Bagi Staf Kumpulan Pengurusan & Profesional (Akademik) pada 24 Oktober 2023:

Bidang:
Agrotechnology, Polymeric biomaterial, Membrane technology Nanocomposite, Wastewater treatment, Biomass, Agricultural residue

Perakuan:
Jawatankuasa memperakukan pelantikan Dr. Sumarni Binti Mansur sebagai Pensyarah Universiti DS51 (Kontrak) berdasarkan justifikasi

6-F	NAMA	DR. SUMARNI BINTI MANSUR
		pelantikan yang telah dinyatakan oleh pihak Pusat Pengajian. Pelantikan ini bagi tempoh dua (2) tahun berkuat kuasa pada tarikh melapor diri dengan gaji pokok sebanyak RM 5,855.00. Pusat Pengajian perlu menetapkan KPI yang jelas kepada staf agar dapat membantu meningkatkan prestasi Pusat Pengajian. Staf juga perlu memenuhi KPI dan kriteria pelantikan semula bagi tujuan pertimbangan untuk pelantikan seterusnya. **Mesyuarat JKPTJSM ke-84 bertarikh 10 Oktober 2023.** **Justifikasi Pelantikan:** Calon mempunyai kelayakan akademik Doktor Falsafah dalam bidang **Agroteknologi dan Biomass** yang kritikal bagi program akademik PPTI teknologi biosumber sedangkan teknologi biosumber sedang dalam proses penawaran program akademik pascasiswazah sarjana mod campuran yang dirancang untuk ditawarkan sidang 2023/2024. Terdapat keperluan mendesak untuk melantik calon dibahagian teknologi biosumber untuk tujuan pengajaran, pembelajaran dan penyelidikan. Cadangan lantikan ini atas alasan: 1. Kepakaran dalam bidang **Agroteknologi dan Biomass** dapat memenuhi keperluan tenaga pengajar dalam **kursus biobahan, biokomposit, biopolimer** dan kursus berkaitan. 2. Mempunyai pengalaman dalam membantu penyeliaan pelajar pascasiswazah dan pelajar latihan industri. 3. Mempunyai pengalaman penyelidikan yang luas dan aktif menerbitkan artikel dalam jurnal berimpak tinggi. 4. Mempunyai h-index Scopus 8 dengan 15 penerbitan dan h-index WoS 8 dengan 14 penerbitan. 5. Mempunyai pengalaman penyelidikan sebagai pegawai penyelidik kontrak dan penyelidik pascadoktoral di universiti penyelidikan terkemuka di Malaysia. 6. Mempunyai anugerah sanjungan akademik/penyelidikan dalam beberapa pertandingan inovasi di peringkat kebangsaan. Perakuan: Mesyurat Jawatankuasa Kecil PTJ pada 10 Oktober 2023. Berdasarkan justifikasi lantikan yang tertera di atas dan calon berdedikasi serta mempunyai aspirasi yang tinggi yang dapat dimanfaatkan oleh pusat pengajian dan universiti. Jawatankuasa memperakukan pelantikan baharu Dr. Sumarni Mansur sebagai Pensyarah Universiti Gred DS51 (Kontrak Tak Berwaran) (Warganegara) bagi tempoh dua (2) tahun berkuatkuasa pada 1 November 2023 hingga 30 Oktober 2025
24.	Catatan Tambahan PTJ	-
25.	Perakuan Jawatankuasa Kecil PTJ	Setelah meneliti prestasi perkhidmatan dan menemu duga staf, Jawatankuasa memperakukan pelantikan semula staf akademik secara kontrak (Warganegara) Dr. Sumarni Binti Mansur bagi tempoh satu (1) tahun berkuat kuasa pada 23 November 2025 hingga 22 November 2026.

6-F	NAMA	DR. SUMARNI BINTI MANSUR
		Ini adalah selaras dengan peraturan baharu Universiti – staf akademik kontrak sedia ada diberikan tempoh satu (1) tahun lantikan semula kontrak bagi tujuan memenuhi syarat lantikan tetap.
26.	Perakuan Jawatankuasa Pemilih	

LAMPIRAN G

4. PELANTIKAN FELO ADJUNG

7-G	NAMA	:	DR. QUAMRUL HASSAN
1	Umur	:	15 Mac 1957 (68 tahun)
2	Warganegara	:	Japan
3	Jenis Lantikan	:	Felo Adjung
4	Kelayakan Akademik	:	PhD. in Agricultural Science Kyoto University, Kyoto, Japan 1992 M.Sc. in Biochemistry University of Dhaka, Bangladesh 1982 B.Sc. Honors in Biochemistry University of Dhaka, Bangladesh 1980 Diploma in Microbiology and Biotechnology Osaka University, Osaka, Japan 1987 Diploma in Industrial Biotechnology GBF-Institute, Braunschweig, Germany 1993
5	Majikan / Organisasi Sekarang	:	Osaka University, Osaka, Japan - Sep 2018-Present Japan Society for Halal Science and Technology - Nov 2018-Present Japan Halal Research Institute (Jahari), Kobe, Japan - July 2014-Present
6	Pengalaman Kerja	:	Professor-Specially Appointed, Center for Global Initiatives Osaka University, Osaka, Japan Sep 2018-Present A Founding Member of Board of Directors Japan Society for Halal Science and Technology Nov 2018-Present Founder Chairman of the Board of Directors Japan Halal Research Institute (Jahari), Kobe, Japan July 2014-Present Founder President Serviceology Society Malaysia, Kedah, Malaysia Feb 2018-Aug 2018 Professor of Technology Management, College of Business; Universiti Utara Malaysia, Kedah, Malaysia (UUM): Aug 2014-Aug 2018

7-G	NAMA	:	DR. QUAMRUL HASSAN
			Director of Knowledge Science Research Centre Universiti Utara Malaysia, Kedah, Malaysia (Uum): Mar 2015-Aug 2018 Founder President & Chief Consultant Bioinnovare Co., Ltd., Kobe, Japan Jan 2006- Jul 2014 Research Scholar Florida State University, Department of Food, Nutrition & Exercise Science 2 Florida, USA 2009 – 2014 1) Visiting Professor, School of Materials Science; 2) Research Professor, Center for Strategic Development of Sci. and Tech. Japan Advanced Institute of Science and Technology, Ishikawa, Japan 1997 – 2005 R&D Manager Mandom Corporation, Osaka, Japan 2001 – 2003 R&D Scientist; R&D Manager Procter & Gamble Company, Ohio, Usa/Kobe, Japan 1994 – 2001 Lecturer; Assistant Professor Department of Microbiology University Of Dhaka, Dhaka, Bangladesh 1983 - 1994
7	Bidang Kepakaran	:	1. Professional biochemist/biotechnologist with over 30 years of experience in both research and technology development and innovation management (in Japan and USA). 2. Held the professorship in a Japanese national university for 8 years, and in a Malaysian national university for more than 4 years (both were previous assignments). 3. Managed the R&D new technology projects in a global company (USA based). 4. Extensively experienced in both multi-cultural and Japanese business and academic settings. 5. Experienced in the management of university-industry R&D collaborations in Japan and USA. 6. Invented, co-developed and successfully launched a new-to-the-world health-care product, from original idea and laboratory test to prototyping and field-test. Procter & Gamble Company (P&G) has marketed this product-Febreze Allergen/House Dust Reducer- globally. 7. Authored/published more than 50 articles/papers/patents and holder of 4 US patents.

<table>
<tr><th>7-G</th><th>NAMA</th><th>:</th><th>DR. QUAMRUL HASSAN</th></tr>
<tr><td></td><td></td><td></td><td>8. Presented as an invited speaker in more than 30 international conferences, and chaired more than 10 conference technical sessions.
9. Professional member of the Institute of Food Technologists, Chicago, USA.
10. Founder Chairman of Japan Halal Research Institute (a non-profit organization), Hyogo, Japan.
11. Founder member of the board of directors of Japan Society for Halal Science and Technology.
12. Founder President of Serviceology Society Malaysia (SESMA), UUM, Kedah, Malaysia.</td></tr>
<tr><td>8</td><td>Skop Tugas/ Petunjuk Prestasi Utama</td><td>:</td><td>
<table>
<tr><th>SO</th><th>Butiran peranan</th><th>Sasaran</th></tr>
<tr><td>SO1</td><td>Menyampaikan syarahan umum mengikut keperluan</td><td>4-8 syarahan umum</td></tr>
<tr><td>SO4</td><td>Membantu menjalinkan kerjasama antara PTJ dengan industri (outbound academic staff)

Membantu menjalinkan kerjasama bagi permohonan geran penyelidikan antarabangsa (Jepun)</td><td>4 - kerjasama (outbound academic staff)

1 - kerjasama (geran penyelidikan Jepun)</td></tr>
<tr><td>SO5</td><td>Membantu penjanaan pendapatan PTJ dalam bentuk bengkel/seminar:</td><td>1 - Bengkel Umum</td></tr>
</table>
</td></tr>
<tr><td>9</td><td>Tempoh Lantikan</td><td>:</td><td>Setahun</td></tr>
<tr><td>10</td><td>Kadar Bayaran</td><td>:</td><td>Mengikut kadar garis panduan Universiti
- Honorarium sebanyak RM500.00 sehari dan tidak melebihi RM4,000.00 sebulan.

- Lantikan Bukan Warganegara dengan bayaran faedah berikut sekali dalam satu lantikan:

o Perjalanan Udara – Tambang perjalanan termurah dan terdekat (tertakluk kepada kelulusan & peruntukan Universiti)
o Penginapan – Dibiayai untuk 8 hari 7 malam (tertakluk kepada kelulusan & peruntukan Universiti)</td></tr>
<tr><td>11</td><td>Peruntukan</td><td>:</td><td>Universiti</td></tr>
<tr><td>12</td><td>Justifikasi Pelantikan</td><td>:</td><td>Lantikan adalah untuk memenuhi keperluan skop tugas/petunjuk prestasi utama, antaranya adalah:
- menyampaikan kuliah/syarahan dalam bidang kepakaran mengikut keperluan
- terlibat dengan pengajaran dan pembelajaran yang berkaitan
- sokongan dalam mendapatkan pengiktirafan dari badan-badan profesional bagi program-program yang berkaitan.
- Sokongan dalam menjalinkan kerjasama dengan industri berkaitan serta masyarakat bagi peringkat nasional dan antarabangsa.</td></tr>
</table>

7-G	NAMA	:	DR. QUAMRUL HASSAN
			- memberi khidmat nasihat pembentukan kurikulum dan silibus program pengajian dan membantu penjanaan kewangan dalam bentuk seminar, *workshop* dan *conference*.
13	Perakuan Jawatankuasa Kecil Sumber Manusia PTJ	:	Jawatankuasa Kecil PTJ – Sumber Manusia memperakukan palantikan Dr. Quamrul Hassan sebagai Felo Adjung di Pusat Pengajian Teknologi Industri bagi tempoh 1 tahun di bawah peruntukan Universiti.
14	Catatan Tambahan (*Untuk Kegunaan PTJ*)	:	
15	Catatan (*Untuk Kegunaan Bahagian Sumber Manusia*)	:	
16	Perakuan Naib Canselor	:	